General Chemistry I

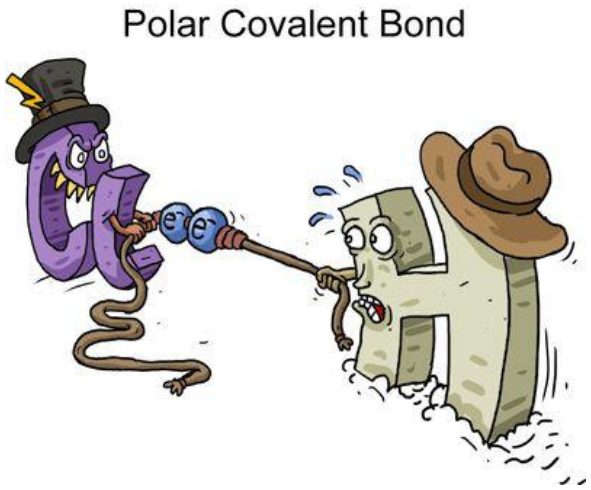
Chemistry: The central science

Phong D. Tran

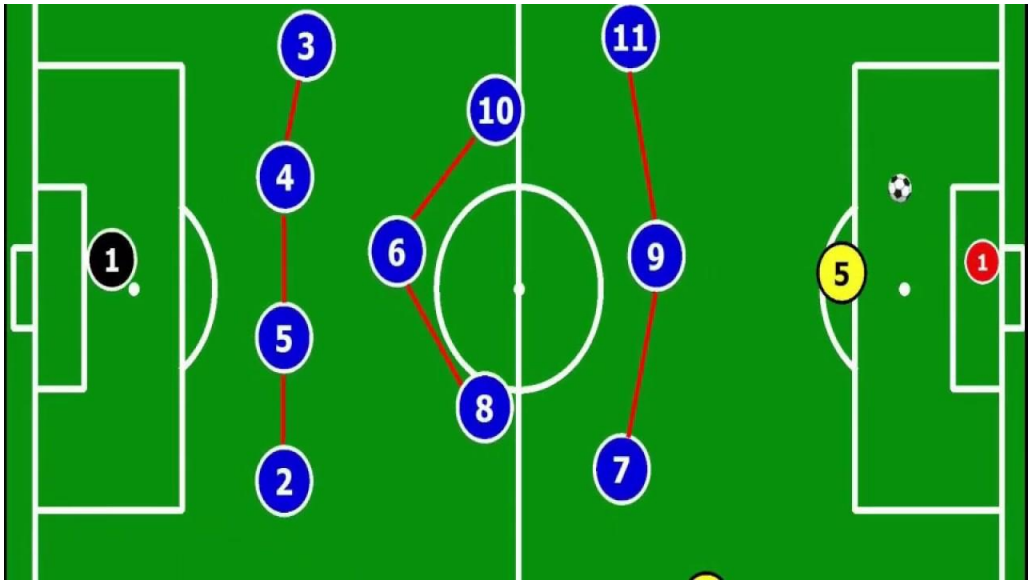
University of Science and Technology of Hanoi

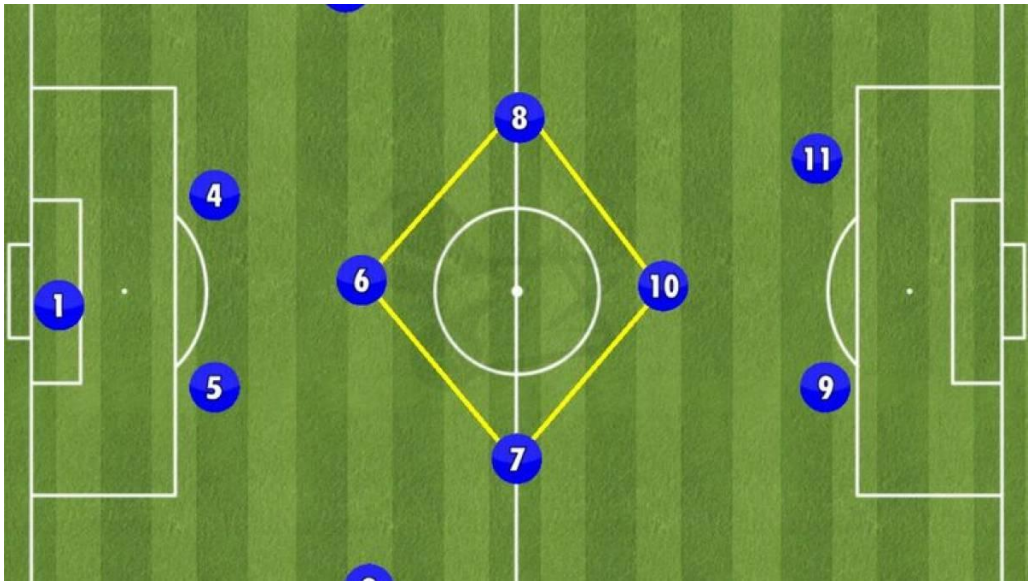
Email: tran-dinh.phong@usth.edu.vn

Lectures 9,10: Basic concept of chemical bonding

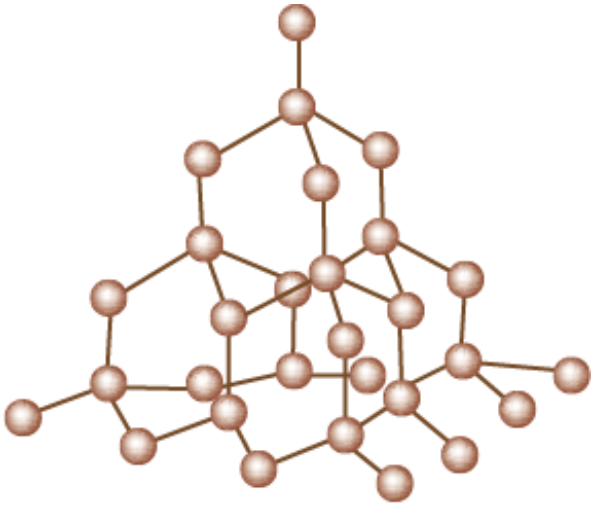


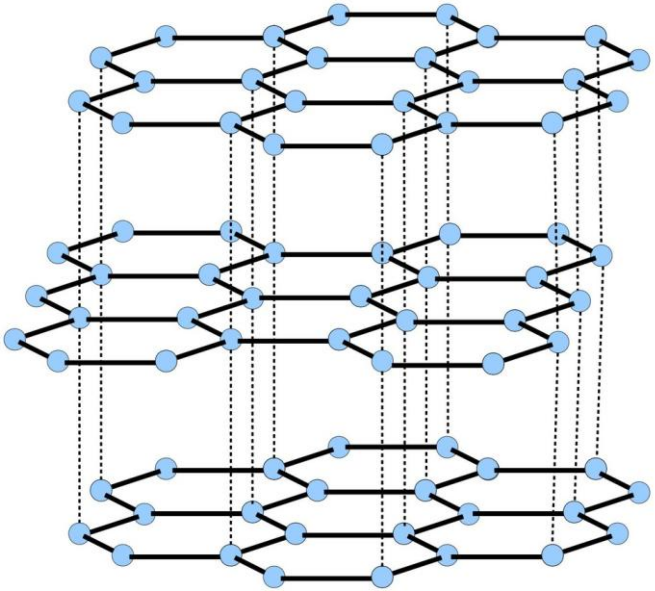
We have already elements, how will we put them together to get properties that we want….???



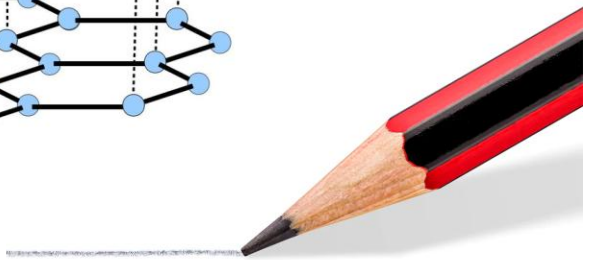


The case of carbon





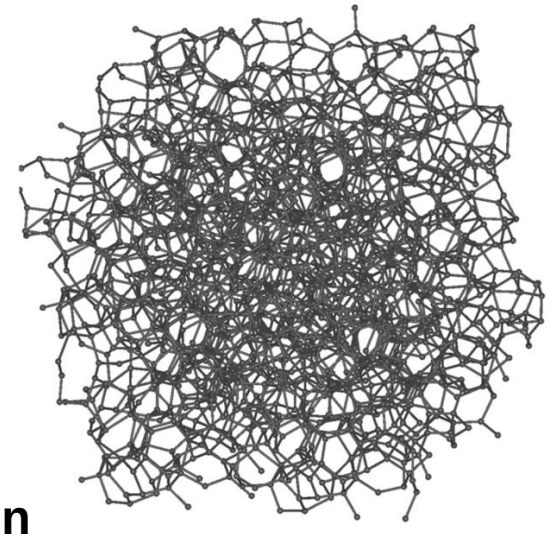
Graphite





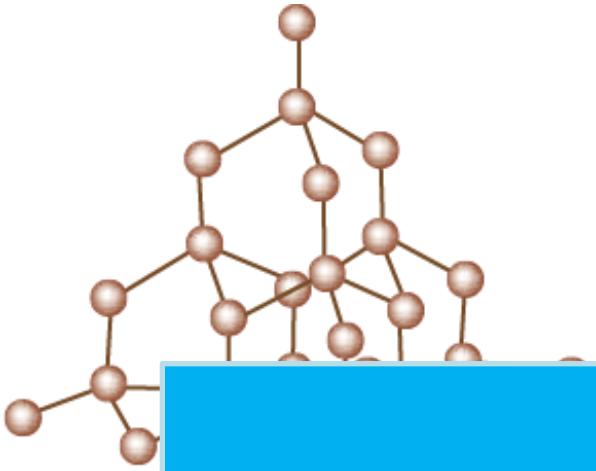
Diamond

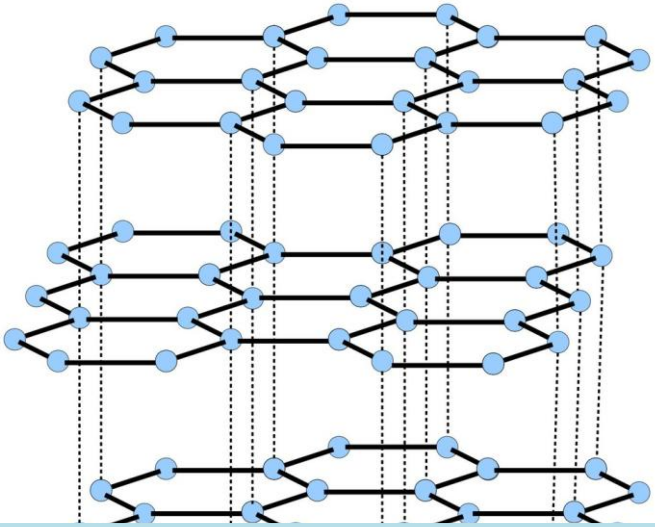




Amorphous Black carbon

The case of carbon





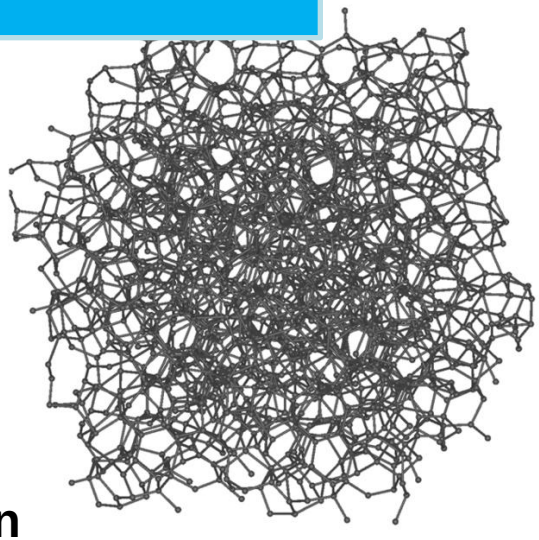


Chemical Bonding



Diamond





Amorphous Black carbon

Content

1. Lewis symbols and the octet rule
2. Ionic bonding
3. Covalent bonding
4. Bond polarity and electronegative
5. Drawing Lewis structure
6. Resonance structure
7. Exceptions to the octet rule
8. Strength and Lengths of covalent bonds

3 types of chemical bonds:

– **Ionic:** NaCl, $NaHCO_3$ (baking soda), KNO_3 (fertilizer)





– **Covalent:** H_2O, CO_2, O_2, Cl_2





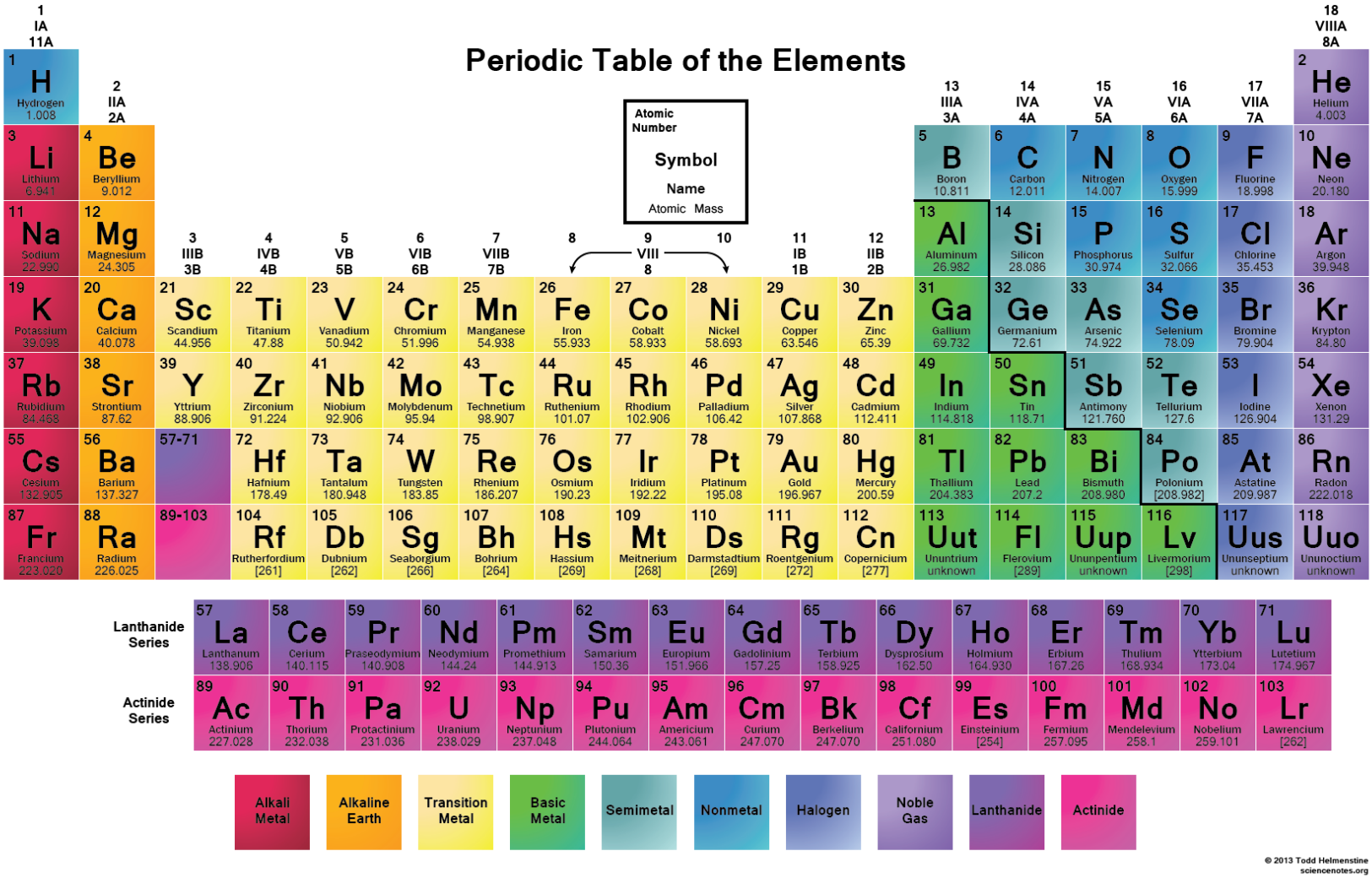
– **Metallic:** Fe, Al, Cu





Group

Row


Periodic Table of the Elements
Atomic Number
Symbol
Name
Atomic Mass
1 IA 11A
2 IIA 2A
3 IIIB 3B
4 IVB 4B
5 VB 5B
6 VIB 6B
7 VIIB 7B
8
9 VIII 8
10
11 IB 1B
12 IIB 2B
13 IIIA 3A
14 IVA 4A
15 VA 5A
16 VIA 6A
17 VIIA 7A
18 VIIIA 8A
1 H Hydrogen 1.008
2 He Helium 4.003
3 Li Lithium 6.941
4 Be Beryllium 9.012
5 B Boron 10.811
6 C Carbon 12.011
7 N Nitrogen 14.007
8 O Oxygen 15.999
9 F Fluorine 18.998
10 Ne Neon 20.180
11 Na Sodium 22.990
12 Mg Magnesium 24.305
13 Al Aluminum 26.982
14 Si Silicon 28.086
15 P Phosphorus 30.974
16 S Sulfur 32.066
17 Cl Chlorine 35.453
18 Ar Argon 39.948
19 K Potassium 39.098
20 Ca Calcium 40.078
21 Sc Scandium 44.956
22 Ti Titanium 47.88
23 V Vanadium 50.942
24 Cr Chromium 51.996
25 Mn Manganese 54.938
26 Fe Iron 55.933
27 Co Cobalt 58.933
28 Ni Nickel 58.693
29 Cu Copper 63.546
30 Zn Zinc 65.39
31 Ga Gallium 69.732
32 Ge Germanium 72.61
33 As Arsenic 74.922
34 Se Selenium 78.09
35 Br Bromine 79.904
36 Kr Krypton 84.80
37 Rb Rubidium 84.468
38 Sr Strontium 87.62
39 Y Yttrium 88.906
40 Zr Zirconium 91.224
41 Nb Niobium 92.906
42 Mo Molybdenum 95.94
43 Tc Technetium 98.907
44 Ru Ruthenium 101.07
45 Rh Rhodium 102.906
46 Pd Palladium 106.42
47 Ag Silver 107.868
48 Cd Cadmium 112.411
49 In Indium 114.818
50 Sn Tin 118.71
51 Sb Antimony 121.760
52 Te Tellurium 127.6
53 I Iodine 126.904
54 Xe Xenon 131.29
55 Cs Cesium 132.905
56 Ba Barium 137.327
57-71
72 Hf Hafnium 178.49
73 Ta Tantalum 180.948
74 W Tungsten 183.85
75 Re Rhenium 186.207
76 Os Osmium 190.23
77 Ir Iridium 192.22
78 Pt Platinum 195.08
79 Au Gold 196.967
80 Hg Mercury 200.59
81 Tl Thallium 204.383
82 Pb Lead 207.2
83 Bi Bismuth 208.980
84 Po Polonium [208.982]
85 At Astatine 209.987
86 Rn Radon 222.018
87 Fr Francium 223.020
88 Ra Radium 226.025
89-103
104 Rf Rutherfordium [261]
105 Db Dubnium [262]
106 Sg Seaborgium [266]
107 Bh Bohrium [264]
108 Hs Hassium [269]
109 Mt Meitnerium [268]
110 Ds Darmstadtium [269]
111 Rg Roentgenium [272]
112 Cn Copernicium [277]
113 Uut Ununtrium unknown
114 Fl Flerovium [289]
115 Uup Ununpentium unknown
116 Lv Livermorium [298]
117 Uus Ununseptium unknown
118 Uuo Ununoctium unknown
Lanthanide Series
57 La Lanthanum 138.906
58 Ce Cerium 140.115
59 Pr Praseodymium 140.908
60 Nd Neodymium 144.24
61 Pm Promethium 144.913
62 Sm Samarium 150.36
63 Eu Europium 151.966
64 Gd Gadolinium 157.25
65 Tb Terbium 158.925
66 Dy Dysprosium 162.50
67 Ho Holmium 164.930
68 Er Erbium 167.26
69 Tm Thulium 168.934
70 Yb Ytterbium 173.04
71 Lu Lutetium 174.967
Actinide Series
89 Ac Actinium 227.028
90 Th Thorium 232.038
91 Pa Protactinium 231.036
92 U Uranium 238.029
93 Np Neptunium 237.048
94 Pu Plutonium 244.064
95 Am Americium 243.061
96 Cm Curium 247.070
97 Bk Berkelium 247.070
98 Cf Californium 251.080
99 Es Einsteinium [254]
100 Fm Fermium 257.095
101 Md Mendelevium 258.1
102 No Nobelium 259.101
103 Lr Lawrencium [262]
Alkali Metal
Alkaline Earth
Transition Metal
Basic Metal
Semimetal
Nonmetal
Halogen
Noble Gas
Lanthanide
Actinide
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sciencenotes.org

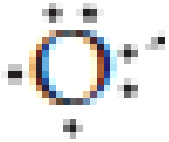
Lewis symbol and the octet rule

- **Valence electron**

  S: [Ne]$\mathbf{3s^2 3p^4}$ O:[He]$\mathbf{2s^2 2p^4}$

- Lewis symbols



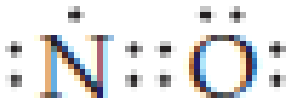


Lewis symbol and the octet rule

Row (period) 2 elements:

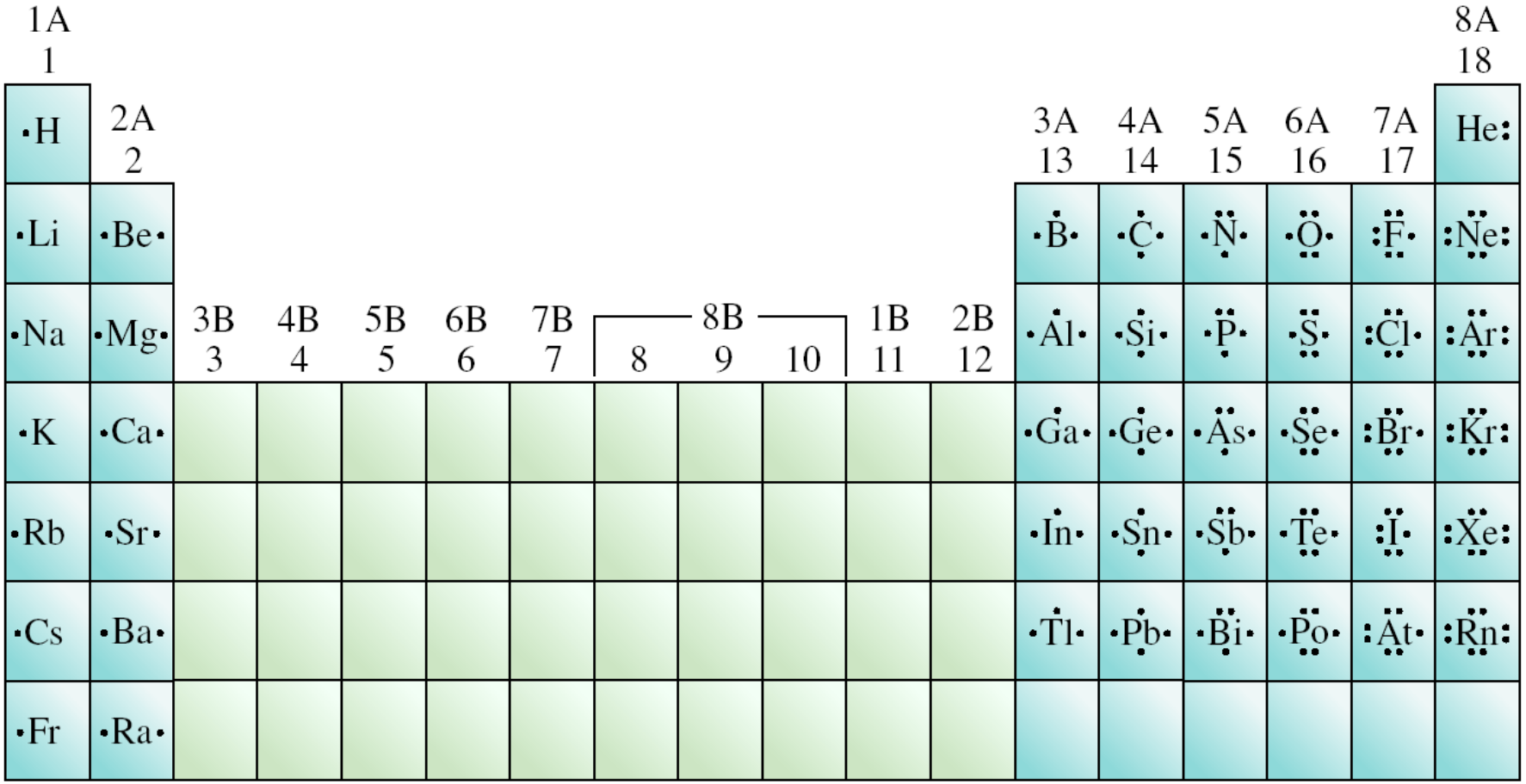


<u>The octet rule</u>: *Atoms tend to gain, lose or share electrons until they are surrounded by eight valence electrons*

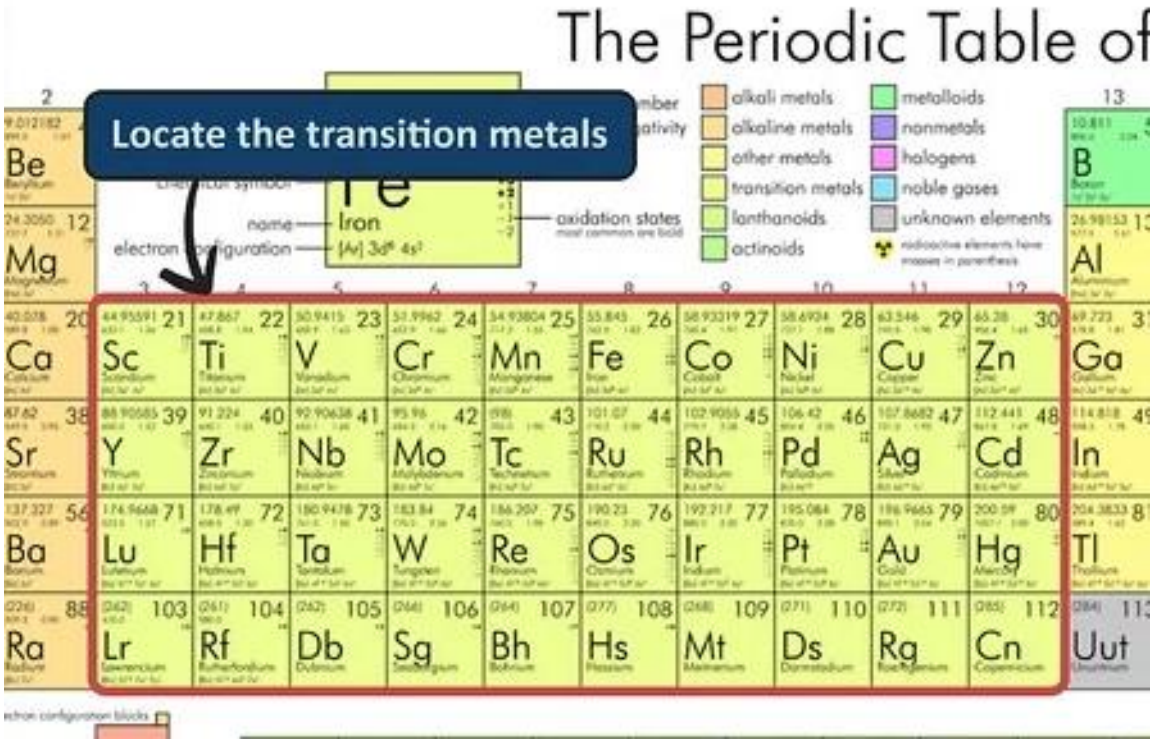
Exception: H and Li, NO 

Lewis symbol and the octet rule

Most useful for main-group elements (s, p elements)



How are about transition metals (d elements)?



Atomic Orbitals of d elements

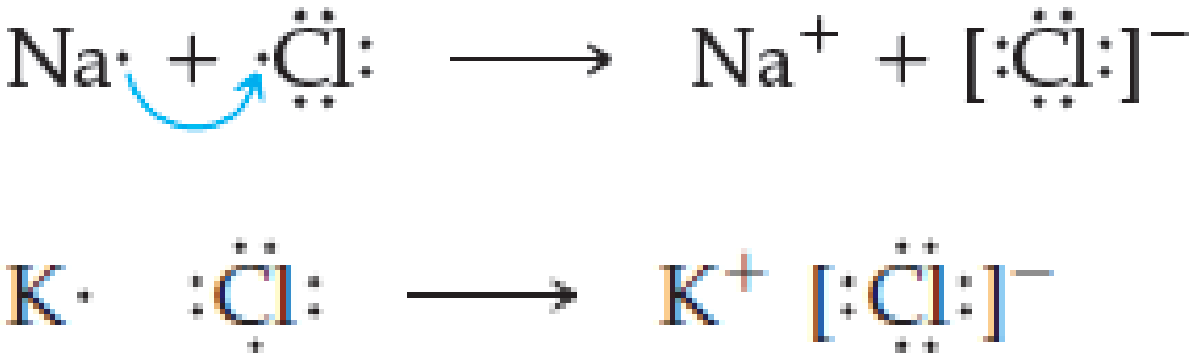
1 AO s

3 AO p

5 AO d

18 electron instead of 8 electron (octet rule)

Ionic Bonding



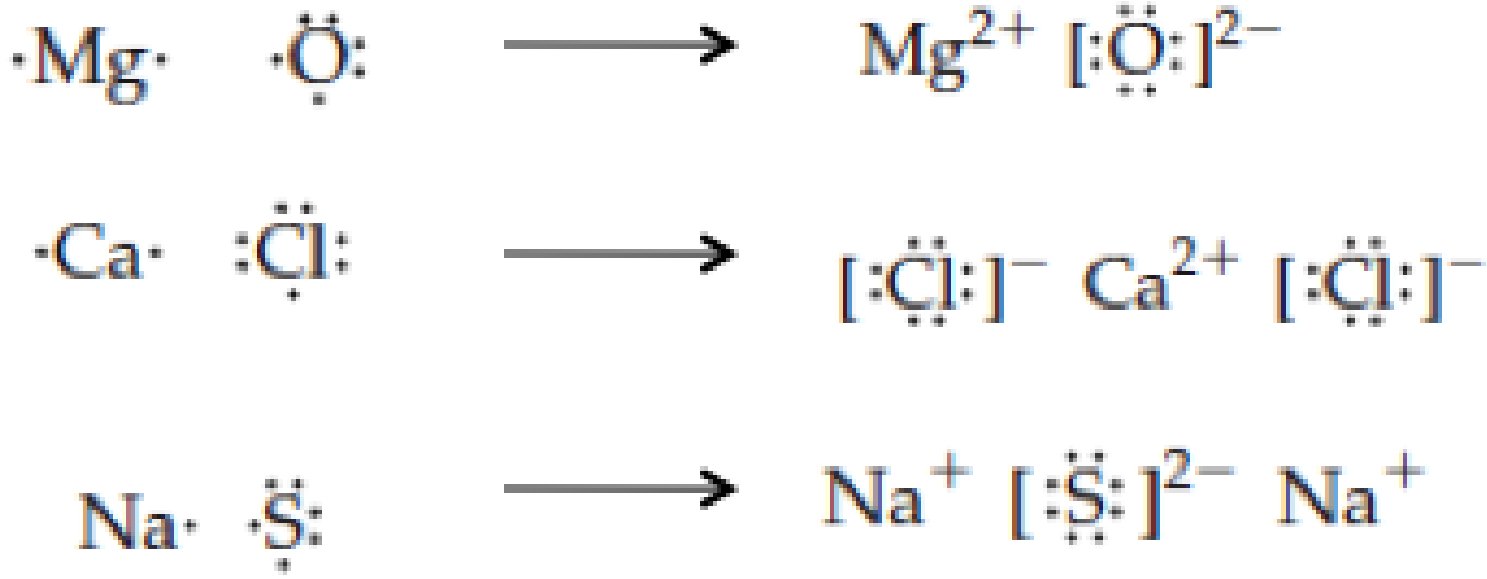
The octet rule -> Ionic bonding

Atoms tend to gain, lose ~~or share~~ electrons until they are surrounded by eight valence electrons

Ionic Bonding

The octet rule -> Ionic bonding

Atoms tend to gain, lose ~~or share~~ electrons until they are surrounded by eight valence electrons



Metals on the left of the periodic table + Non-metals on the right of the periodic table

Ionic bonding

- Sodium chloride formation experiment

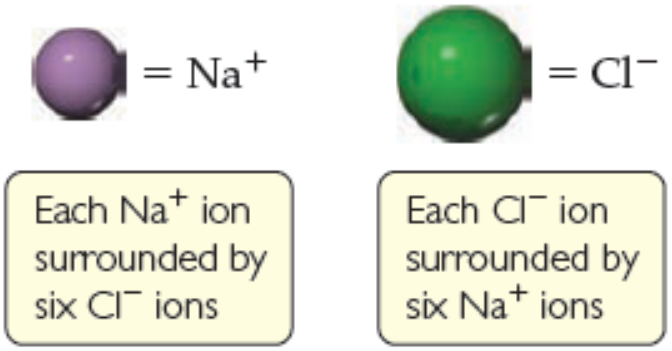


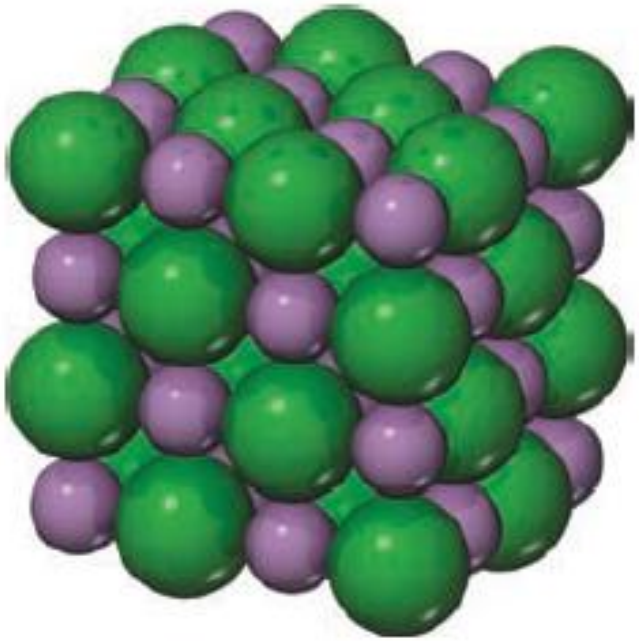
https://youtu.be/d2geiGKFveE

Ionic bonding

- **Why ionic bonding is formed?**
  - Metal: ready to gives up electron(s)
  - Non-metal: high electron affinity

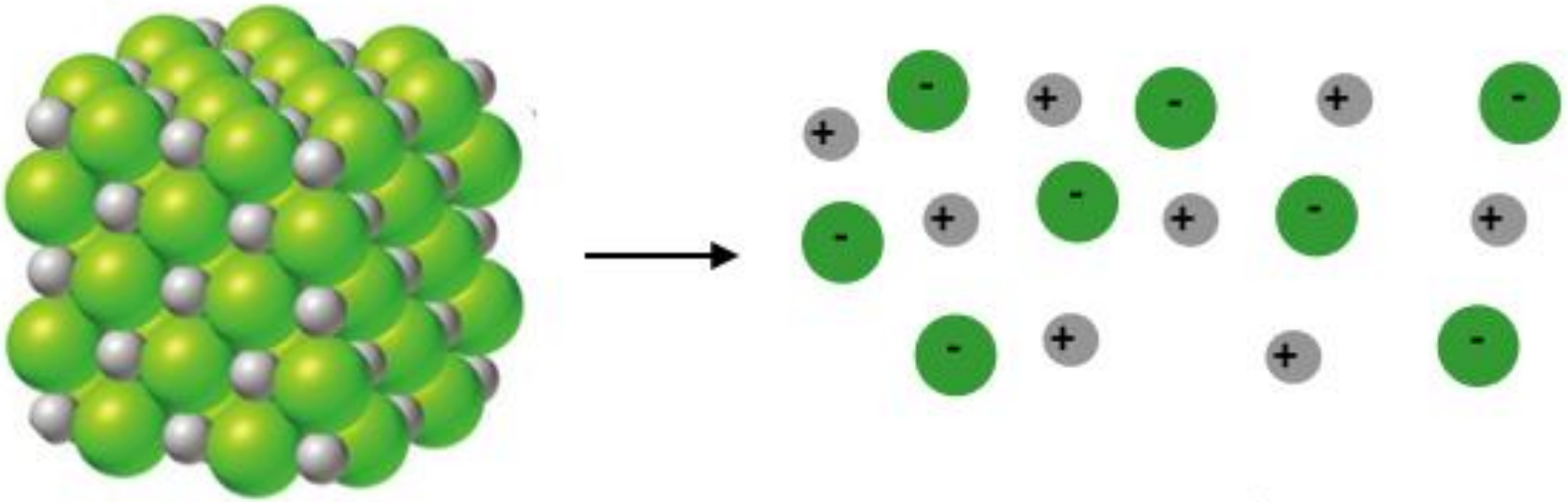
  -> electron transfer from metal to non-metal
- **Ionic bonding characteristics:**
  - Hard and brittle
  - high melting points
  - crystalline
  - break apart along smooth, flat surfaces
- Ionic bonding formation is **exothermic** due attraction between oppositely charged ions





Lattice energy

- Lattice energy: Energy required to completely separate one mole of a solid ionic compound into its gaseous ions



Always positive

$$NaCl(s) \longrightarrow Na^+(g) + Cl^-(g) \qquad \Delta H_{lattice} = +788\ kJ/mol$$

Comparison: Energy to boil 500 ml of water: ~ 170 kJ

Lattice energy

Potential energy of 2 interacting charged particles

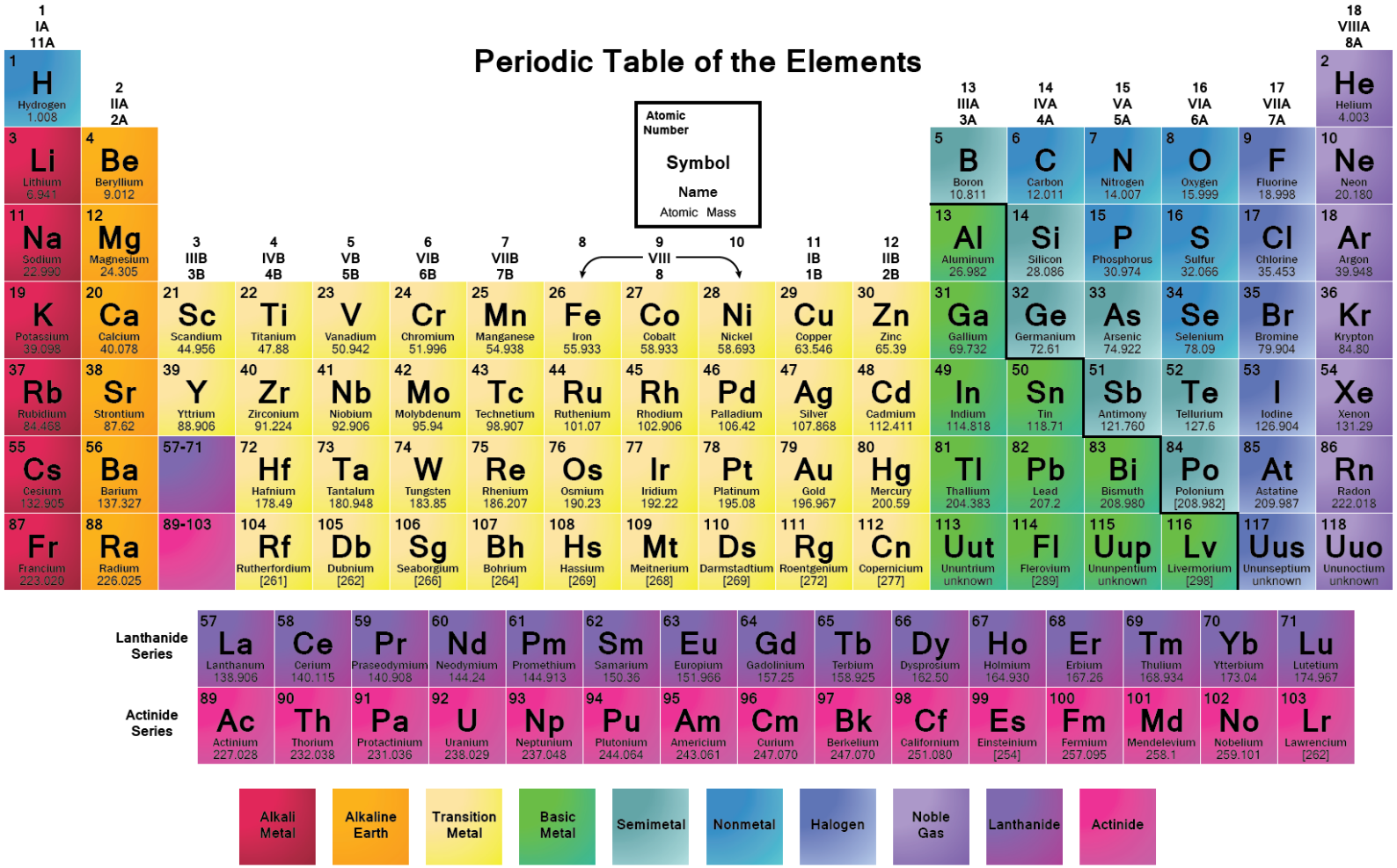
$$E_{el} = \frac{\kappa Q_1 Q_2}{d}$$

For a given arrangement of ions, the lattice energy increases as:

- the charges on the ions increase
- the radii decrease

Example

Arange the following ionic compounds in order of increasing lattice energy: NaF, CsI, and CaO



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Increasing lattice energy: NaF, CsI, CaO

A) NaF < CsI < CaO

B) CsI < NaF < CaO

C) CaO < CsI < NaF

$$E_{el} = \frac{\kappa Q_1 Q_2}{d}$$

Please place following compounds in order of increasing lattice energy:

KBr, KF, $MgCl_2$ and MgO

$$E_{el} = \frac{\kappa Q_1 Q_2}{d}$$

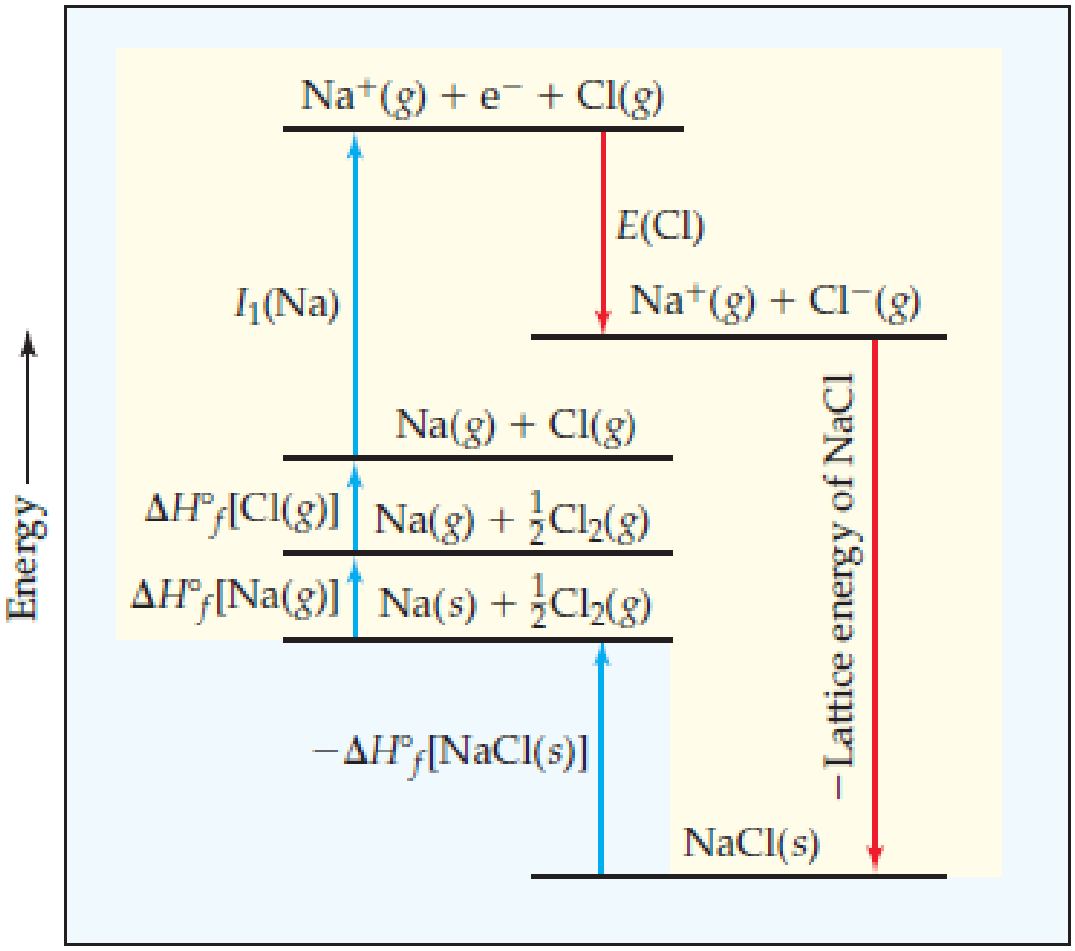
Ionic bonding

Table 8.2 Lattice Energies for Some Ionic Compounds

Compound	Lattice Energy (kJ/mol)	Compound	Lattice Energy (kJ/mol)
LiF	1030	$MgCl_2$	2326
LiCl	834	$SrCl_2$	2127
LiI	730		
NaF	910	MgO	3795
NaCl	788	CaO	3414
NaBr	732	SrO	3217
NaI	682		
KF	808	ScN	7547
KCl	701		
KBr	671		
CsCl	657		
CsI	600		

Born-Harber cycle: Calculate lattice energy

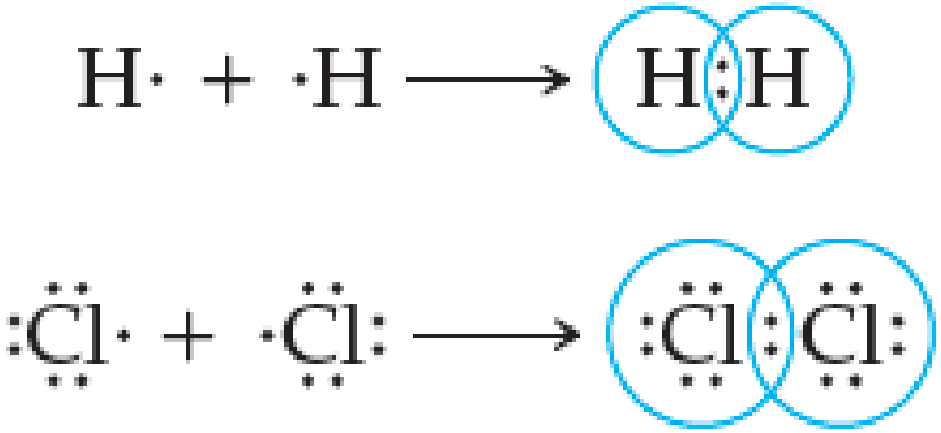
Based on Hess's law: Total enthalpy of a process = summation of enthalpy of all of its stages



Covalent bonding

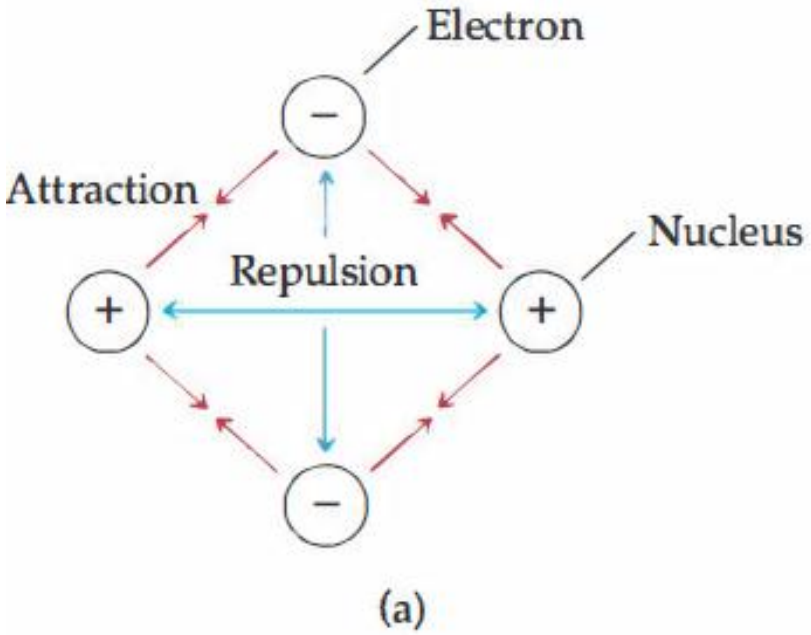
- The octet rule -> Covalent bonding

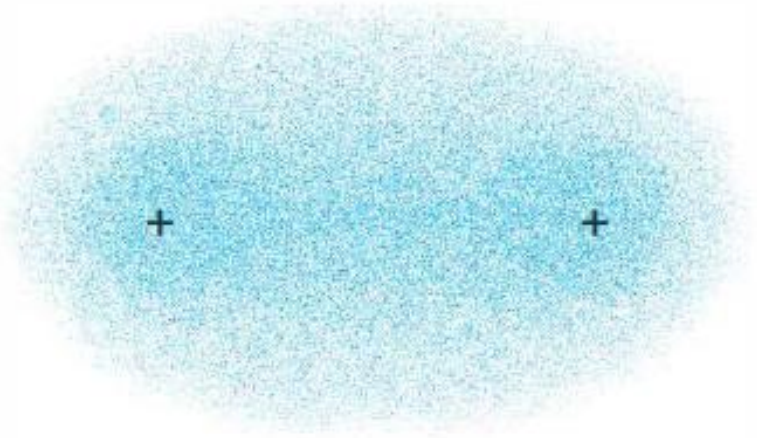
Atoms tend to ~~gain, lose or~~ share electrons until they are surrounded by eight valence electrons



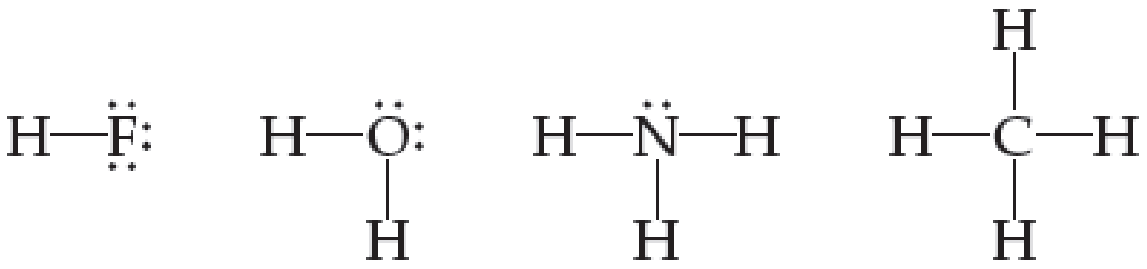
Why octet rule ($8e^-$)?

Covalent bond





Types of covalent bond



Single bond

(Bond order = 1)



Double bond

(Bond order = 2)



Triple bond

(Bond order = 3)

Possible bond order of **0.5, 1.5?**

Bond strength and bond length

bond strength single < double < triple

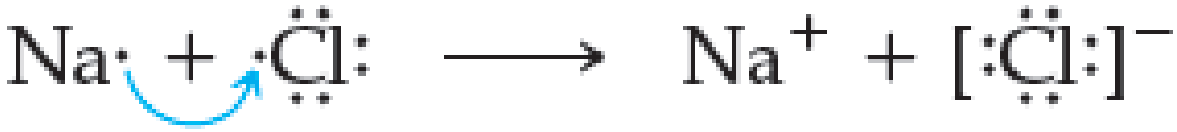
bond length single > double > triple

	N – N	**N = N**	**N ≡ N**
Bond Strength	163 kJ/mol	418 kJ/mol	941 kJ/mol
Bond Length	1.47 Å	1.24 Å	1.10 Å

Why this evolution?

Bond polarity

Ionic bonding



Covalent bonding



Bond polarity: measurement of the equality of the shared electrons

Nonpolar covalent bond: electrons are shared equally

Polar covalent bond: one atom attracts the sharing electrons more than the other atom

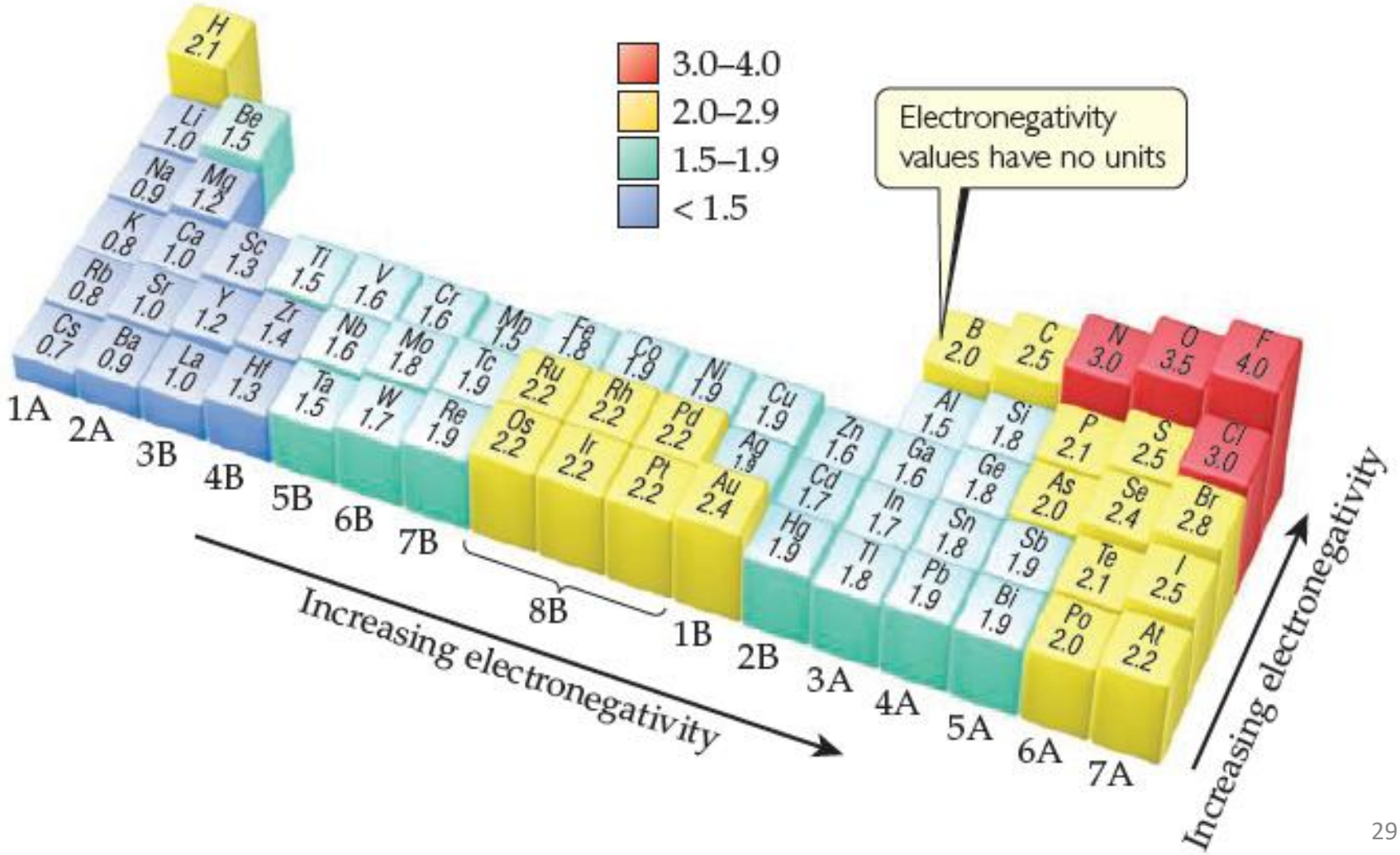
Bond polarity

M:X	$M^{\delta+}X^{\delta-}$	$M^{+}X^{-}$
Pure covalent bond	**Polar covalent bond**	**Ionic bond**
Neutral atoms held together by *equally* shared electrons	Partially charged atoms held together by *unequally* shared electrons	Oppositely charged ions held together by electrostatic attraction

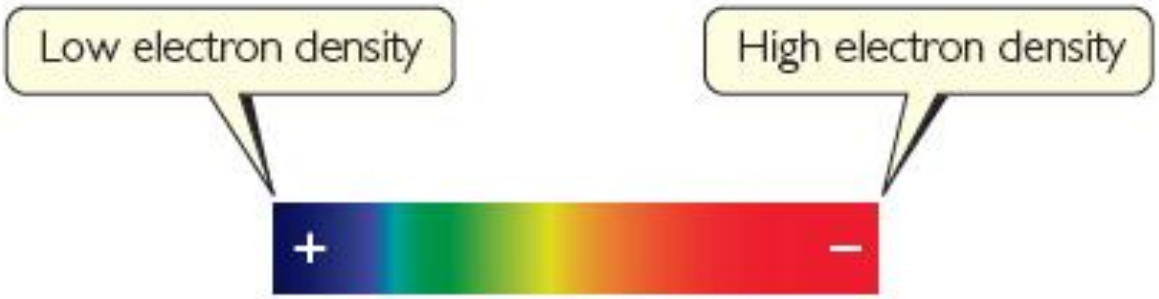
How could we tell if a given bond is polar covalent bond, nonpolar covalent bond or ionic?

Electronegativity

- **Electronegativity**: ability of an atom to attract electron to itself



Electronegativity and Bond Polarity





F_2 HF LiF

Non-polar covalent **Polar covalent** **Ionic**

How to quantify a molecule polarity?

Dipole moment

- **Dipole**: two electrical charges of equal magnitude and opposite sign

  e.g. HCl, LiF

- **Dipole moment**: Magnitude of a dipole

Dipole moment

$$\mu = Qr$$

Dipole moment

Unit: debyes (D)

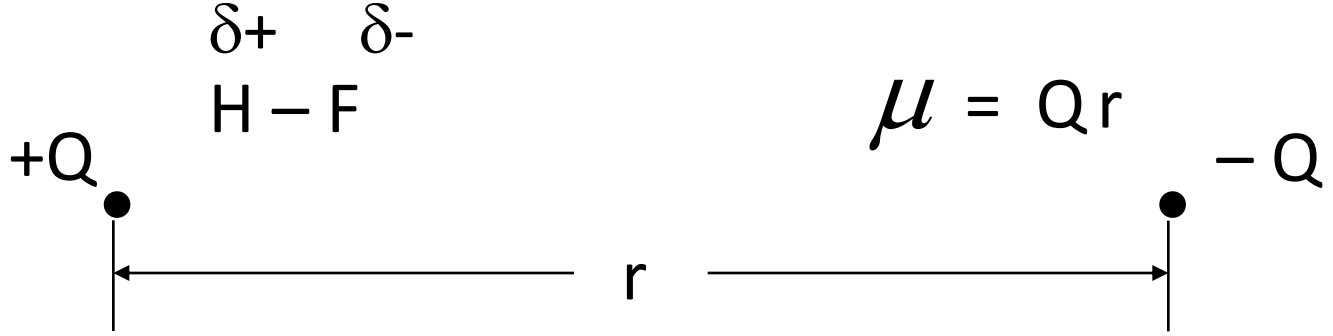
Electrical charge

Unit: Coulombs (C)

Distance of the two atoms

Unit: meters (m)

1 debyes (D) = 3.34 x 10^{-30} coulomb.meter (C.m)



Dipole moment example

The bond length in the HCl molecule is 1.27 A^{o}

a) Calculate dipole moment, assume the charges are H: +1, Cl: -1

b) The measured dipole moment of HCl(*g*) is 1.08 D, calculate the charges on H and Cl atoms

1 elementary charge = $1.6\ 10^{-19}$ C

Dipole moment example

a) Use dipole moment equation $\mu = Qr$

The charges on H and Cl atoms are +1 and -1, Q = 1.6×10^{-19} C

r = 1.27A^{o} = 1.27×10^{-10}m

$$\mu = Qr = (1.60 \times 10^{-19} C)(1.27 \times 10^{-10} m) \frac{1D}{3.34 \times 10^{-30} C.m} = 6.08D$$

Dipole moment example

μ = 1.08D, r = 1.27 Å. Calculate Q

$$Q = \frac{\mu}{r} = \frac{1.08D\,\frac{3.34\times10^{-30}C.m}{1D}}{1.27\times10^{-10}} = 2.84\times10^{-20}C$$

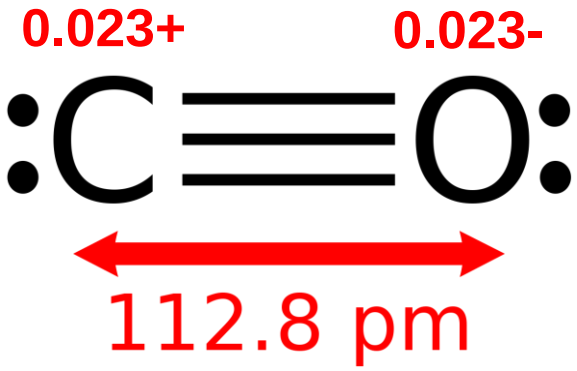
Convert the charge to electron charge:

$$2.84\times10^{-20}\,\frac{1e}{1.6\times10^{-19}C} = 0.178e$$

$$\overset{0.178+}{\mathrm{H}} - \overset{0.178-}{\mathrm{Cl}}$$

Conclusion: HCl bond in HCl(g) is polar covalent (not ionic)

Dipole moment example



$$\mu = Qr$$

μ = **0.122 D** Calculate Q?

1 debyes (D) = 3.34 x 10^{-30} coulomb.meter (C.m)

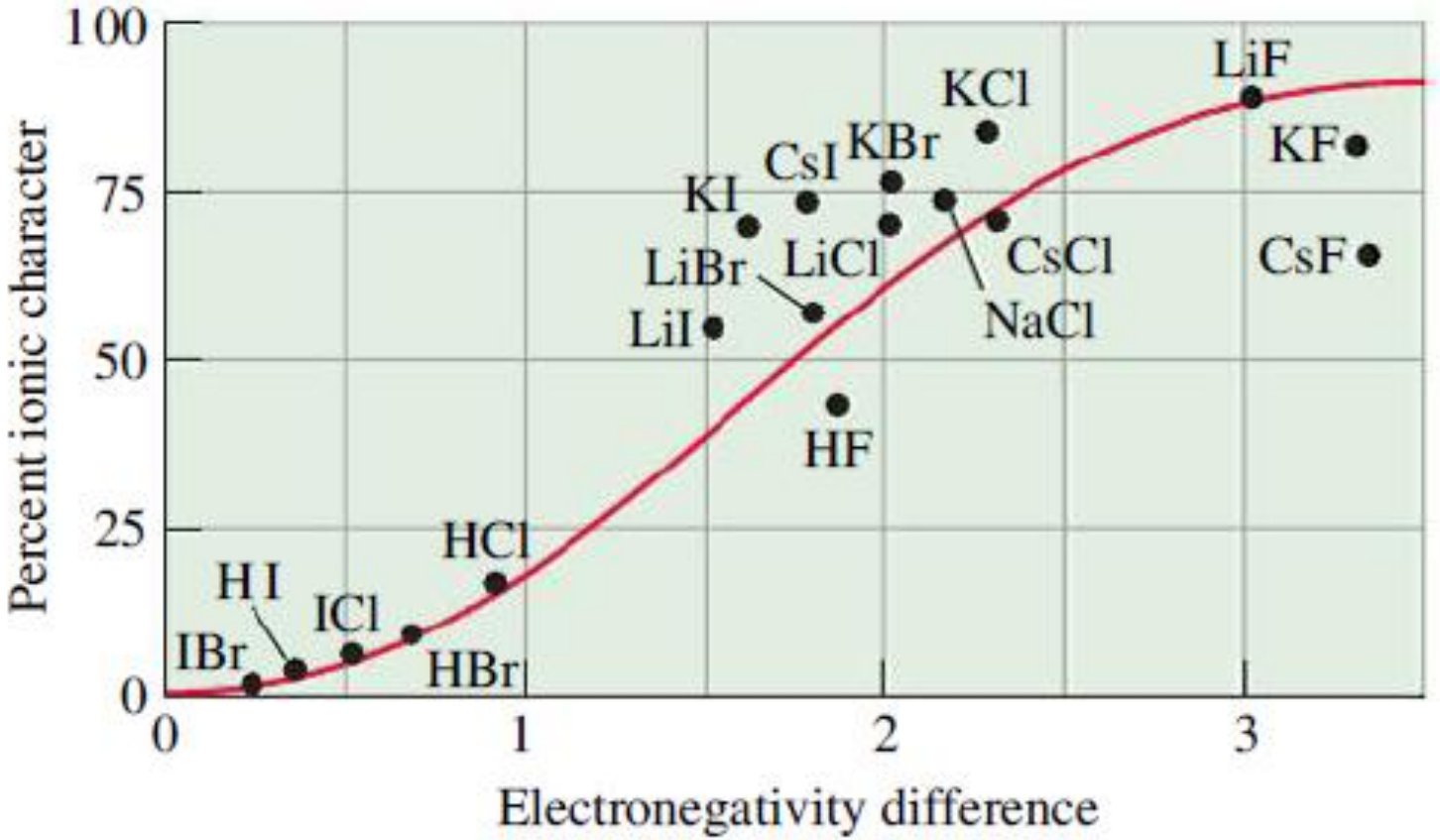
1 elementary charge = 1.6 10^{-19} C

Electronegativity and Bond Polarity

Ionic

Polar covalent

Non-polar covalent



$$\Delta EN = |EN_A - EN_B|$$

Dipole moment example

HCl	ΔEN = 3.0 - 2.1 = 0.9	μ = 1.08D
CO	ΔEN = 3.5 - 2.5 = 1.0	μ = 0.122D

How can we explain this phenomenon?

Key points

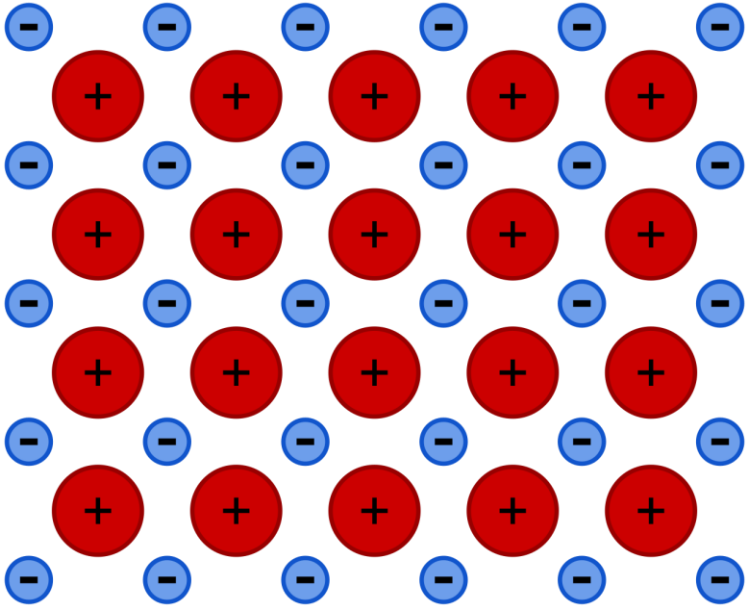
1. Lewis symbols to draw valence electron and the octet rule (8 valence electrons)
2. Ionic bonding (atoms gain or lose electrons)
3. Covalent bonding (atoms share electrons)
4. Bond polarity and electronegativity

Metallic Bond

Metallic Bond

Metallic bonding: Electrostatic attractive force between conduction electrons (in the form of an electron cloud of delocalized electrons) and positively charged metal ions.

It can be also described as the *sharing of free electrons* among a *structure of positively charged ions* (Cations)



Metallic Bond Strength:

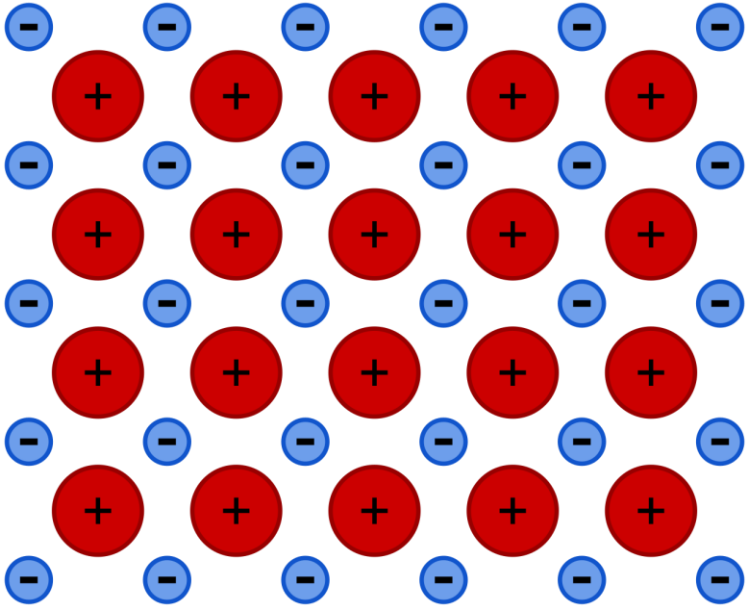
↑ Number of valence electron ↑ strength

↑ Atomic radius ↓ strength

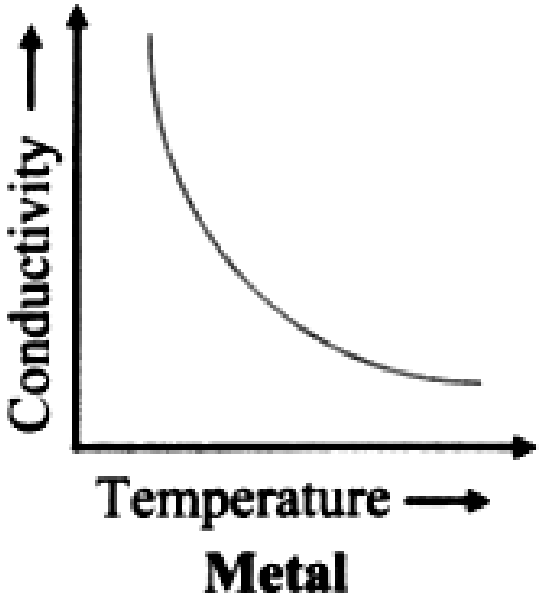
Metallic Bond

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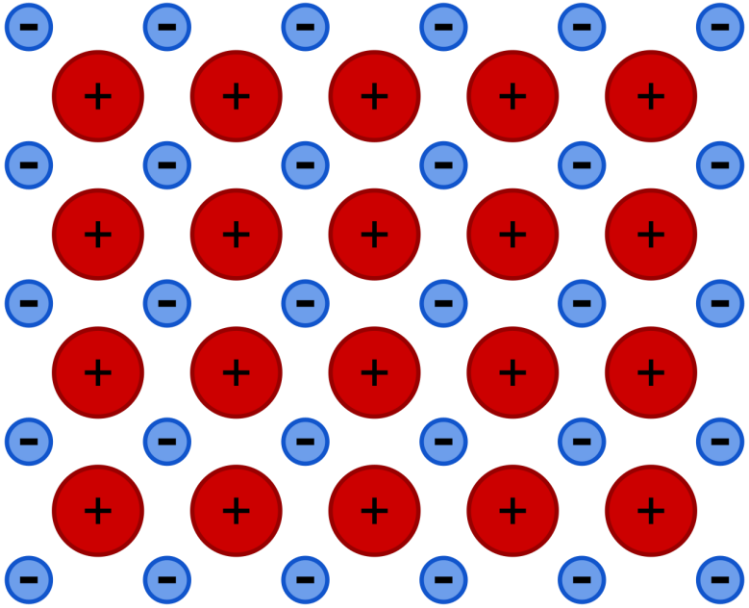


Electrical conductivity of metal *vs.* temperature

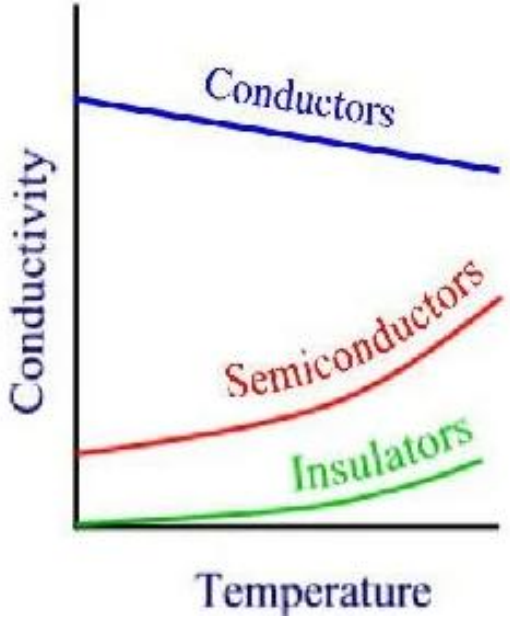


Metallic Bond

Metallic bonding: Electrostatic attractive force between conduction electrons (in the form of an electron cloud of delocalized electrons) and positively charged metal ions. It can be also described as the *sharing of free electrons* among a *structure of positively charged ions* (Cations)



Electrical conductivity of matter *vs.* temperature



Drawing Lewis structure

1. Sum the valence electrons from all atoms
2. Write the symbols for the atoms, the element with the least electronegativity is in the center, and connect them with a single bond
3. Complete the octets around all the atoms bonded to the central atom
4. Place any leftover electrons on the central atom
5. If there are not enough electrons to give the central atom an octet, try multiple bonds

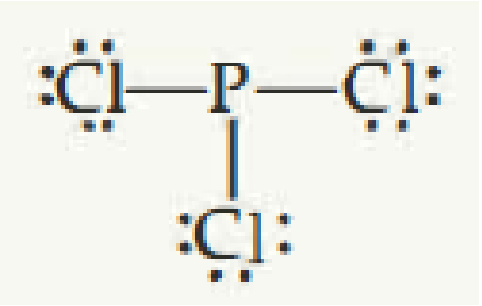
Drawing Lewis structure

Draw Lewis structure for PCl_3

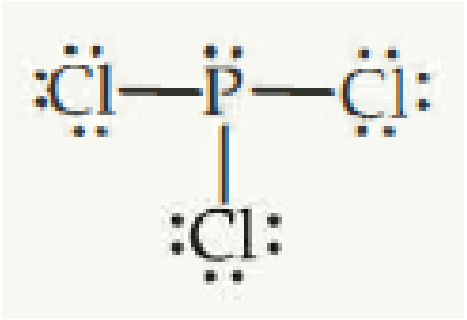
1. Sum the valence electrons: 5 + (3x7) = 26
2. Arrange the atoms: Cl – P – Cl
   |
   Cl

(The atom with less electronegativity in the center)

3. Complete the octet in Cl:



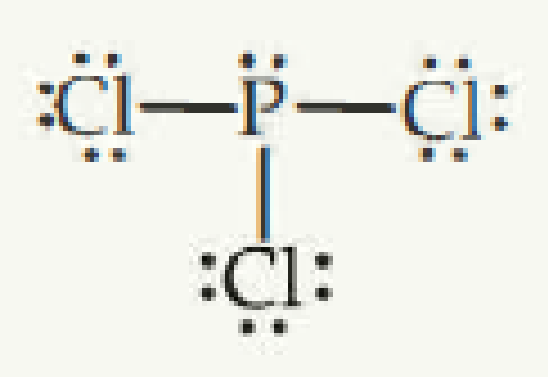
4. Complete the octet in P:



5. Each atom has an octet, we can stop here

Lewis acid/ base

Is PCl_3 a Lewis acid or a Lewis base?



An Acid-Base reaction:

$Ni^{2+} + 4\ PCl_3 \rightarrow [Ni(PCl_3)]^{2+}$

Drawing Lewis structure

NO_2

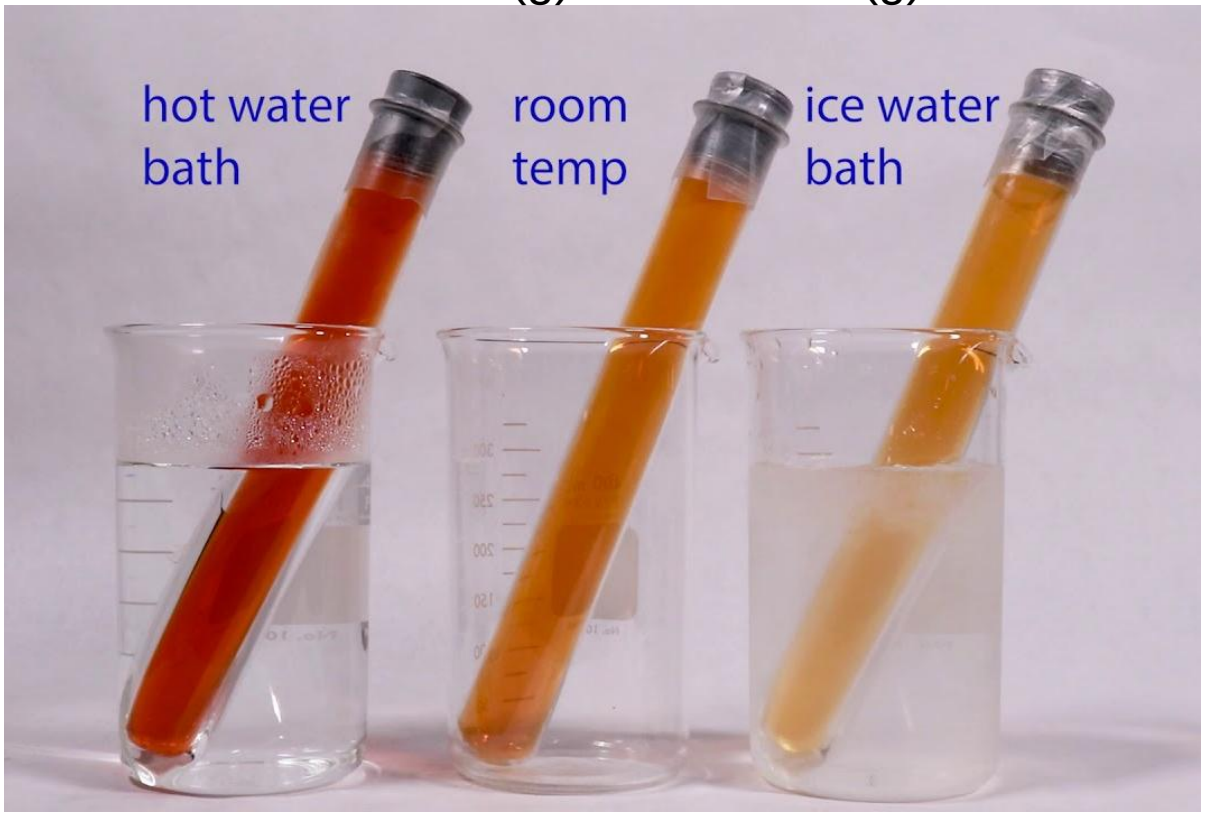
How would NO_2 act (do) to satisfy the octet rule for N atom?

Drawing Lewis structure

NO_2

How would NO_2 act (do) to satisfy the octet rule for N atom?

$$2NO_{2(g)} \rightleftarrows N_2O_{4(g)}$$



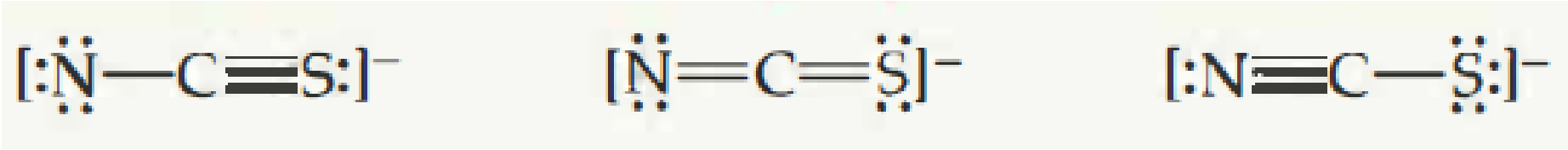
Drawing Lewis structure

NO_2

More Practice:

Draw Lewis structure for BrO_3^- ion and HCN

Formal charge

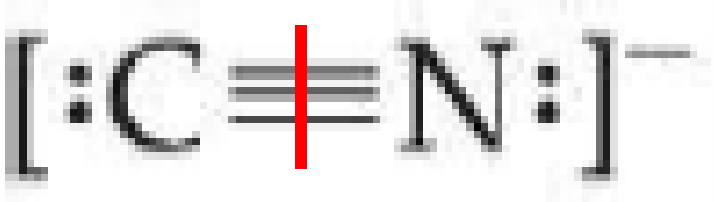


When there are more than one way to draw Lewis structure, to decide which way is correct, we must use formal charge

Formal charge

The charge the atom would have if all the atoms in the molecule had the same electronegativity (each bonding electron pair in the molecule were shared equally between its two atoms)

-1 **0**



- C has 4 valence electrons
- N has 5 valence electrons

5 assigned electron

5 assigned electron

Determine the formal charge on these Lewis structures



(1)



(2)



(3)

To choose the correct structure:

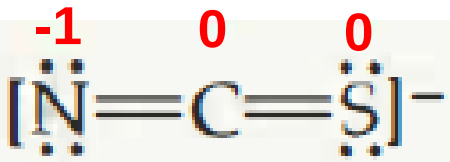
1. We generally choose the Lewis structure in which the atoms bear formal charges closest to zero.

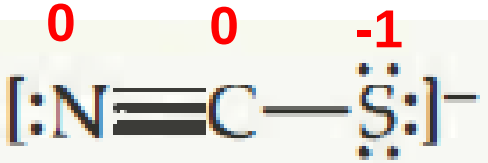
2. We generally choose the Lewis structure in which any negative charges reside on the more electronegative atoms.

<u>Remember</u>: formal charge is not real charge of the atoms, it is just a way to decide which Lewis structure is correct

Determine the formal charge on these Lewis structures







To choose the correct structure:

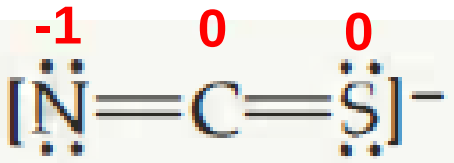
1. We generally choose the Lewis structure in which the atoms bear formal charges closest to zero.

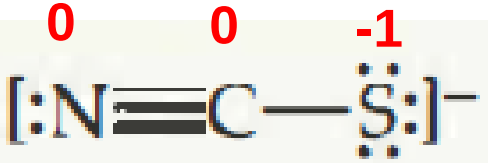
2. We generally choose the Lewis structure in which any negative charges reside on the more electronegative atoms.

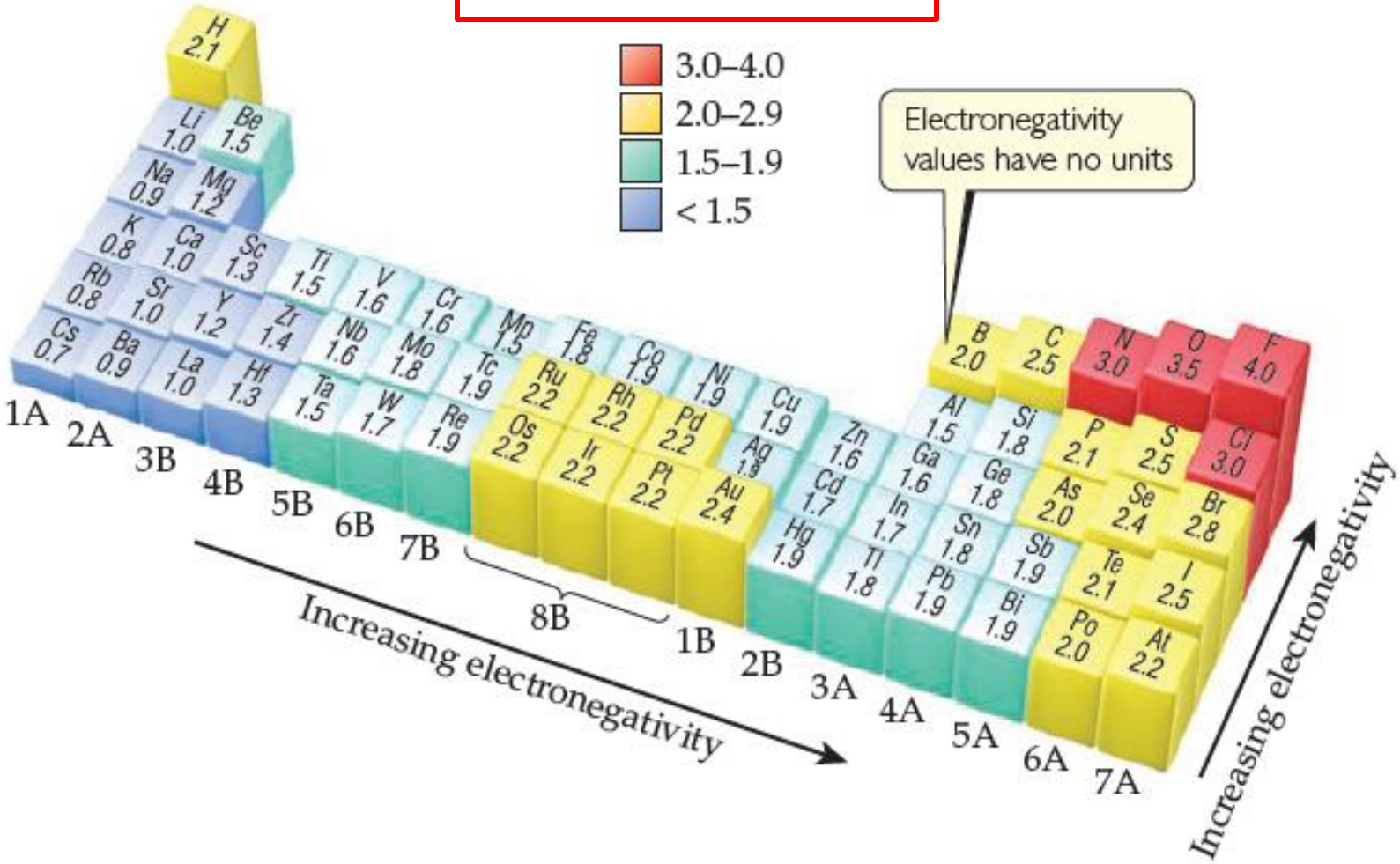
<u>Remember</u>: formal charge is not real charge of the atoms, it is just a way to decide which Lewis structure is correct

Determine the formal charge on these Lewis structures



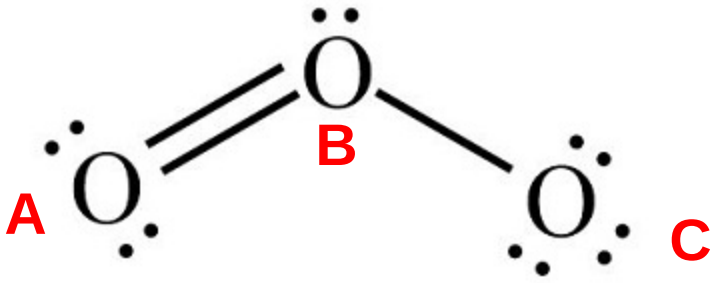






Resonance structures

O_3 molecule

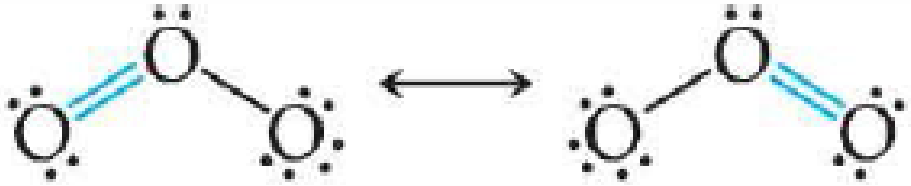


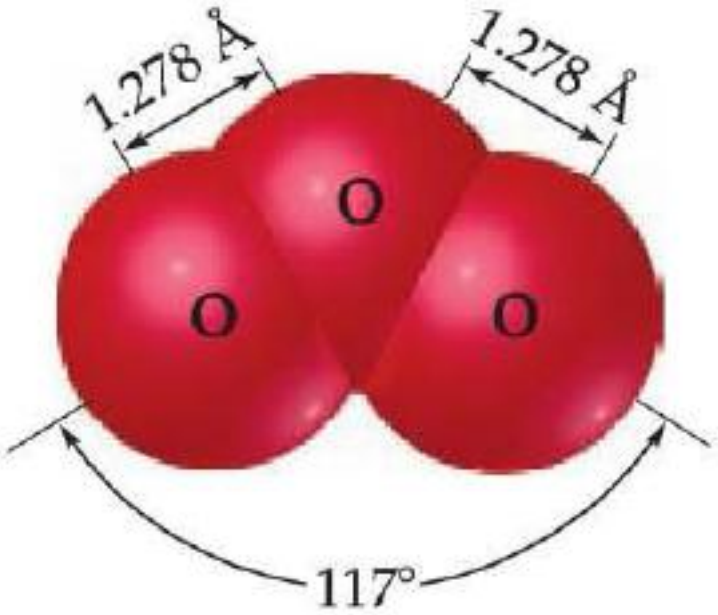
One may draw following conclusions?

$O_{(A)} \neq O_{(C)}$

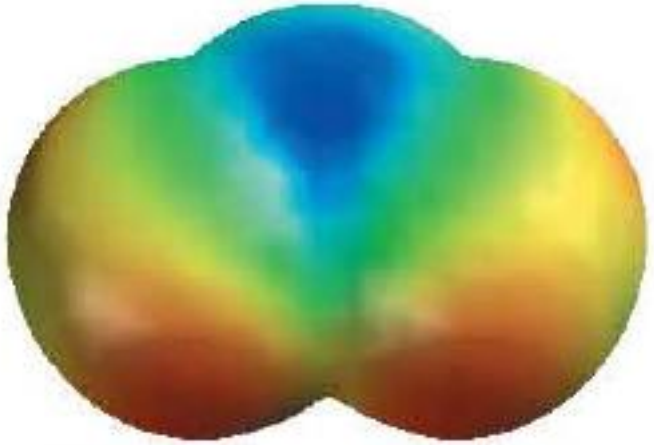
$O_{(A)} = O_{(B)}$ bond is shorter (and stronger) than $O_{(C)} - O_{(B)}$ bond

Resonance structure



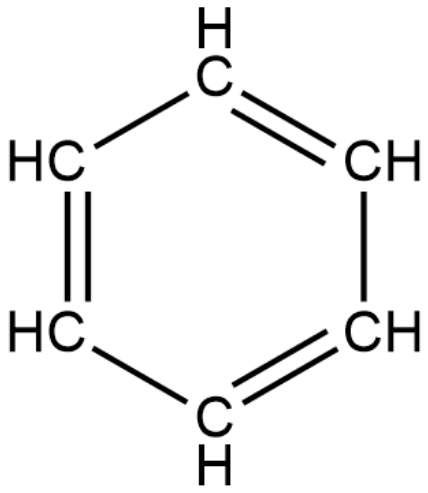


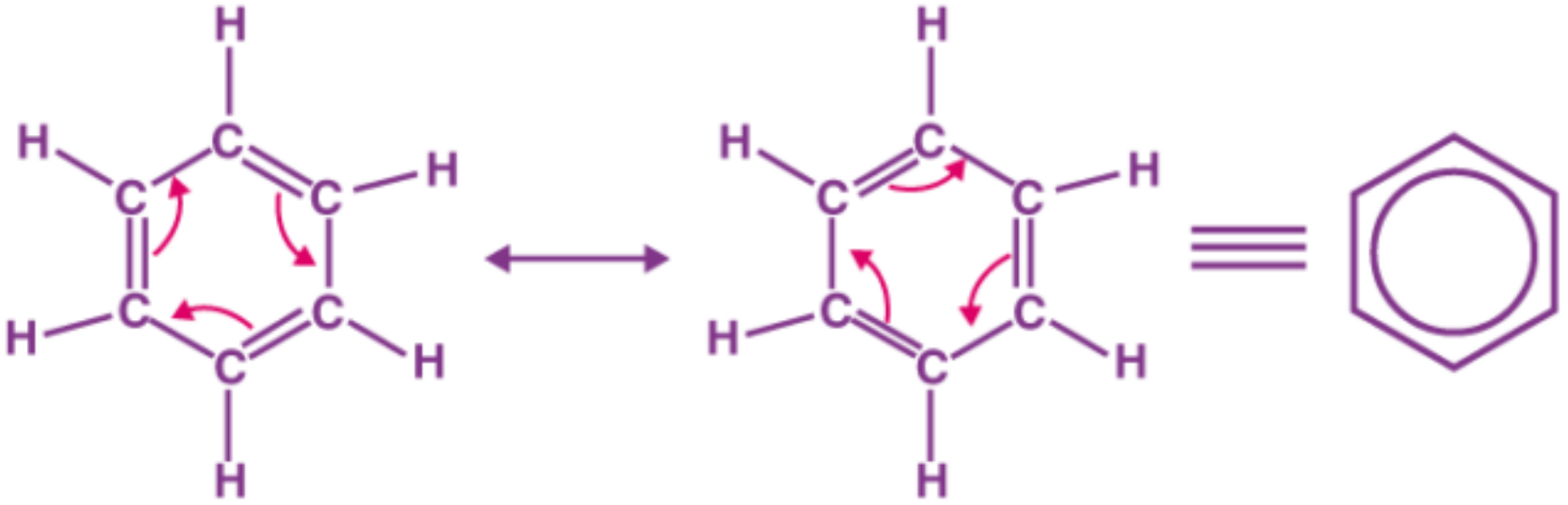
The ozone molecule is a blend of two resonance structures. We cannot describe the ozone molecule in terms of a single Lewis structure.



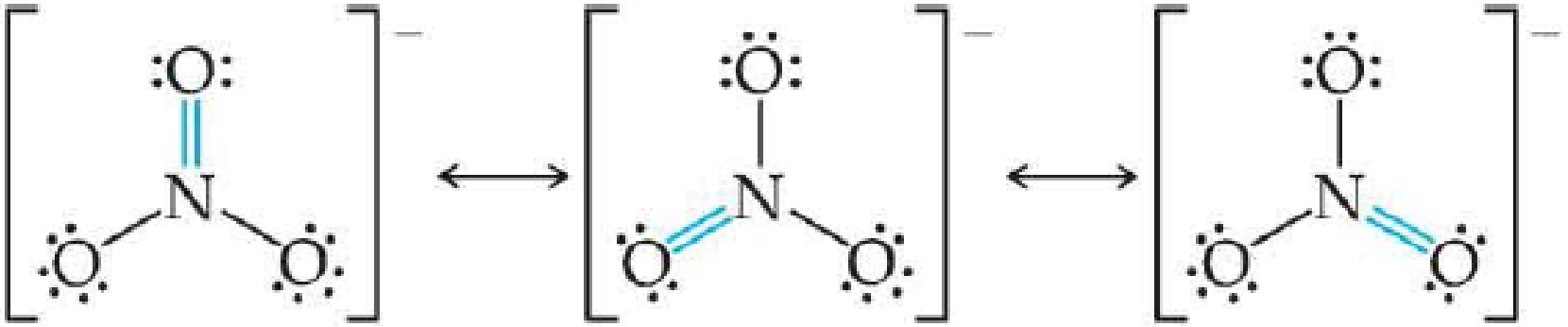
Resonance structures

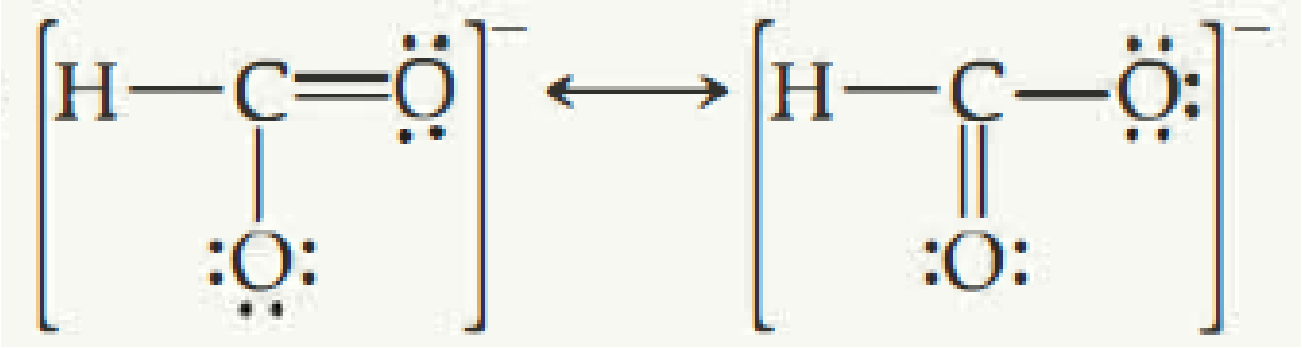
Benzene





Resonance structures





Resonance structures are used when two or more equally valid Lewis structures can be written

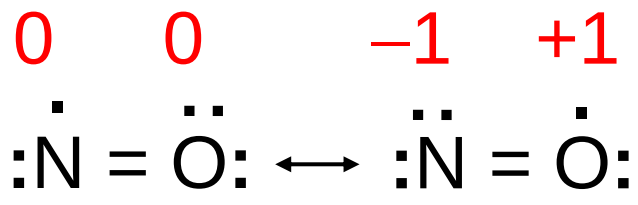
Exceptions to the octet rule

- **Exceptions are of the 3 main types:**
  - Odd number of electrons
  - Fewer than an octet of valence electrons
  - More than an octet of valence electrons

Odd Numbers of Electrons

NO (nitrogen monoxide or nitric oxide)

$$NO \Rightarrow (1 \times 5) + (1 \times 6) = 11 \text{ valence } e^-$$

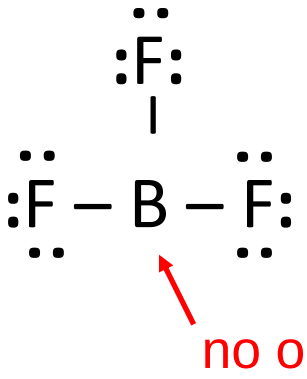


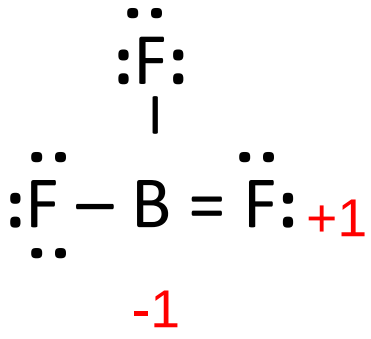
Are these both equally good?

Fewer than an octet

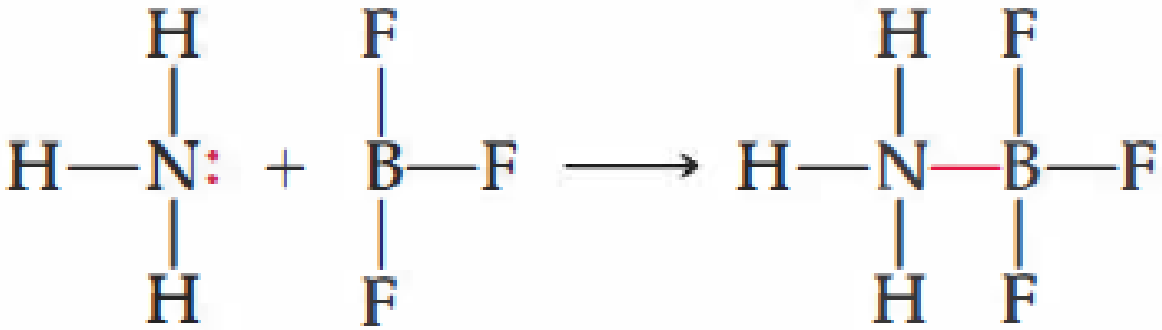
BF_3 (boron trifluoride)

$BF_3 \Rightarrow (1 \times 3) + (3 \times 7) = 24$ val. e^-

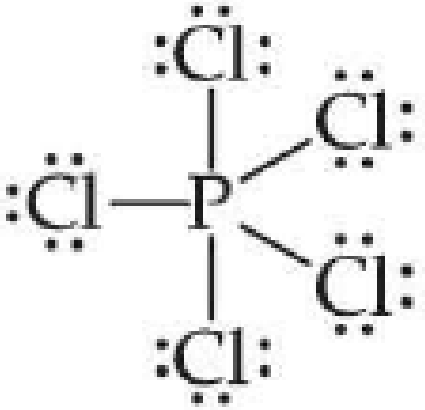


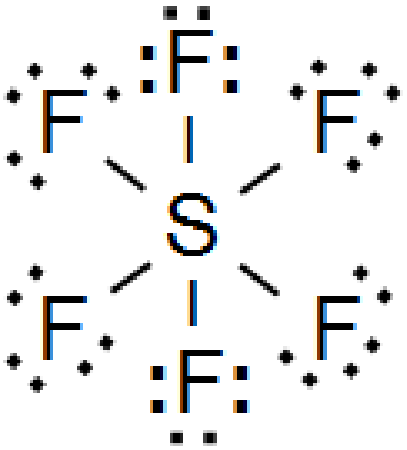


React strongly with NH_3

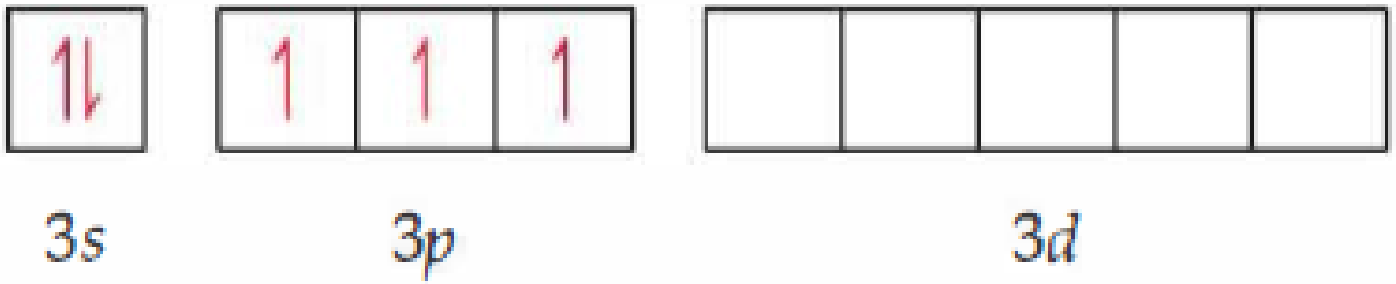


More than an octet



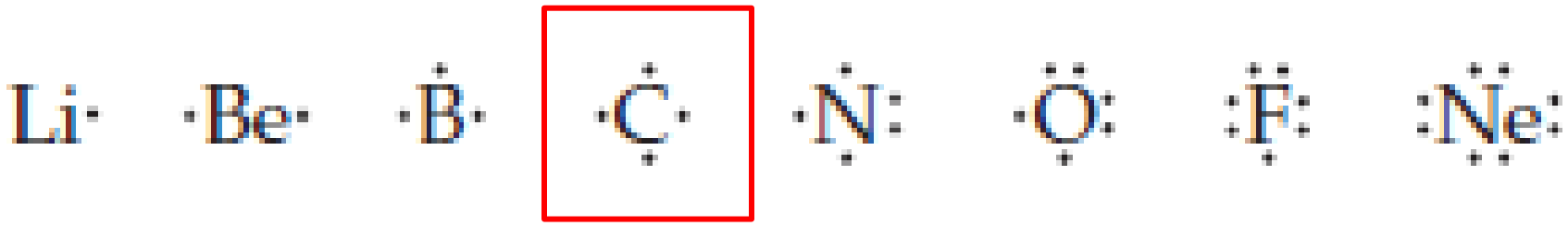


- Elements of the 3rd period and beyond have *d*-orbitals that allow more than 8 valence electrons
  (*rule of 18 electrons for coordination compounds*)



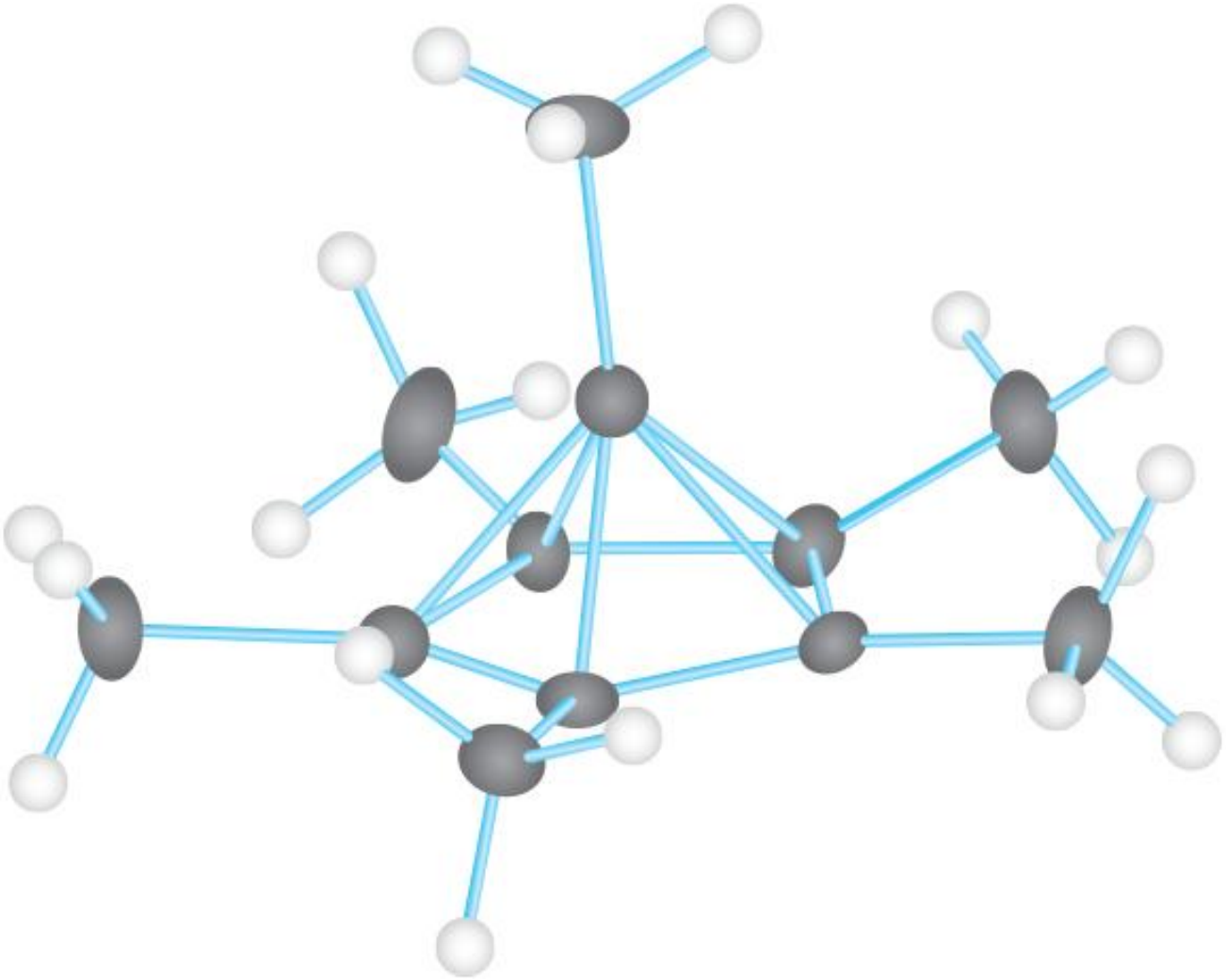
Chemistry of Carbon

Row (period) 2 elements:

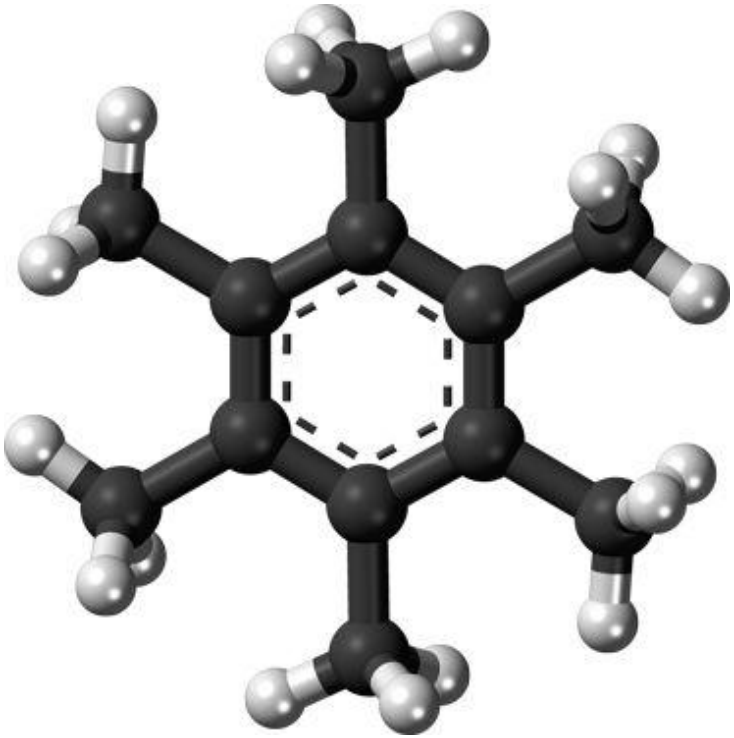


What is the maximal number of covalent bonds that C could form?

Carbon can exceed four-bond limit

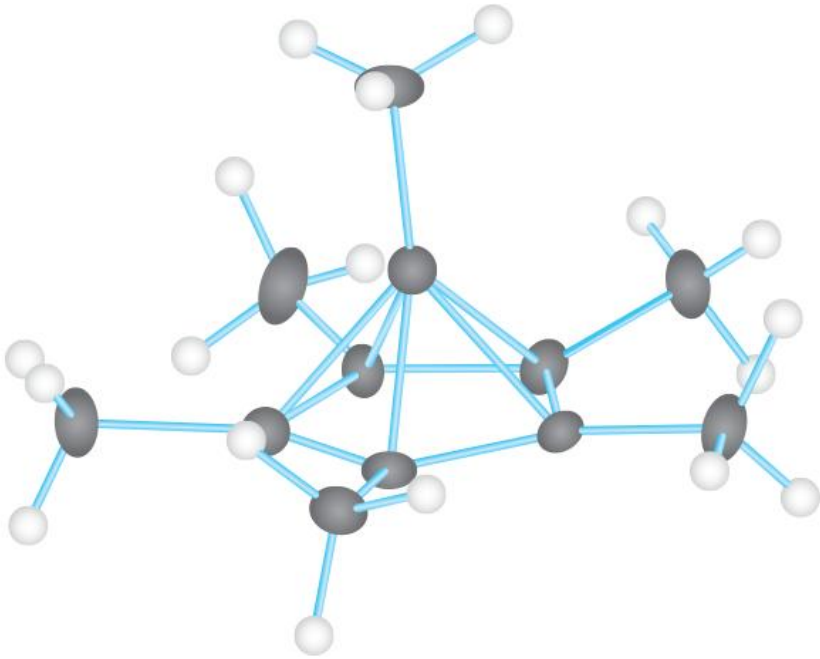


M. Malischewski and K. Seppelt. Crystal structure determination of the pentagonal-pyramidal hexamethylbenzene dication $C_6(CH_3)_6^{2+}$. *Angewandte Chemie*. Vol. 56, **January 2, 2017**, p. 368



$-2e^-$

In strong acid



Strength and length of covalent bonds

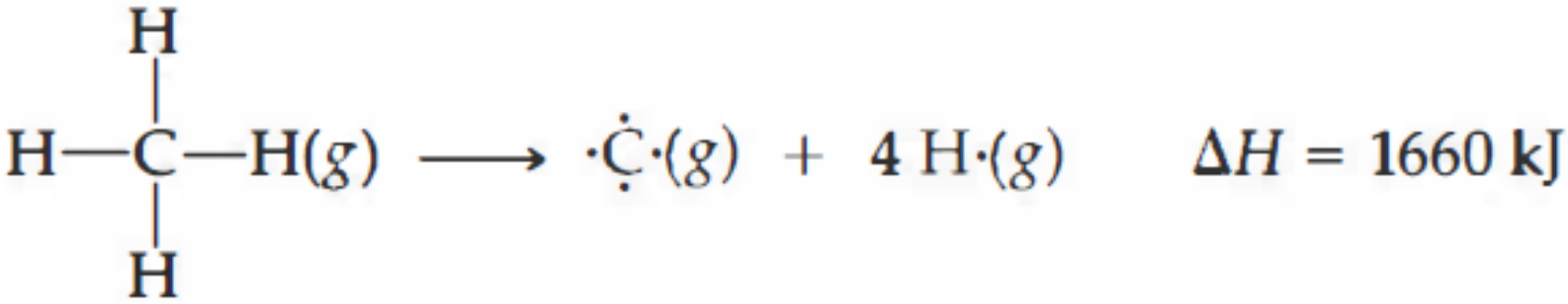
Bond enthalpy: the enthalpy change, ΔH, for the breaking of a particular bond in one mole of a gaseous substance

For diatomic molecules these are accurately measured quantities

$Cl_2(g) \rightarrow Cl(g) + Cl(g)$	$\Delta H^o = 243.4$ kJ	single bonds
$HCl(g) \rightarrow H(g) + Cl(g)$	$\Delta H^o = 431.9$ kJ	
$O_2(g) \rightarrow O(g) + O(g)$	$\Delta H^o = 495.0$ kJ	double bond
$N_2(g) \rightarrow N(g) + N(g)$	$\Delta H^o = 945.4$ kJ	triple bond

Strength of covalent bonds

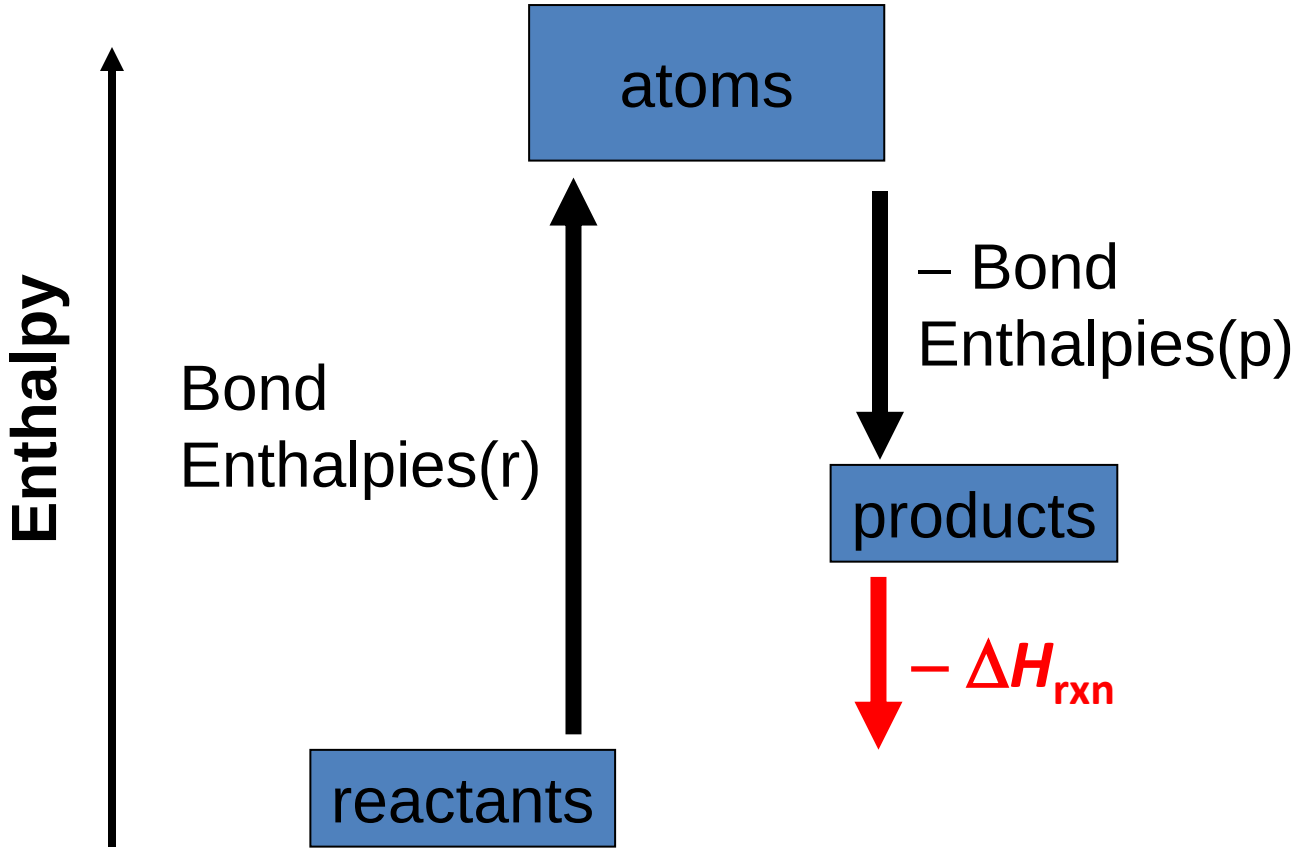
Average bond enthalpies are used for polyatomic molecules



C-H bond enthalpy 1660 kJ/ 4mol = 415 kJ/mol

A molecule with strong chemical bonds generally has less tendency to undergo chemical change than does one with weak bonds.

Enthalpies of reactions



$$\Delta H_{rxn} = \sum \text{Bond Enthalpies(reactants)} - \sum \text{Bond Enthalpies(products)}$$

$\Delta H_{rxn} > 0$ endothermic

$\Delta H_{rxn} < 0$ exothermic

Enthalpies of reactions

$$H{-}CH_3(g) + Cl{-}Cl(g) \longrightarrow Cl{-}CH_3(g) + H{-}Cl(g) \quad \Delta H_{rxn} = ?$$

Reactant bond broken: 1 mol C-H, 1 mol Cl – Cl

Product bond created: 1 mol C-Cl, 1 mol H - Cl

$$\Delta H_{rxn} = [D(C{-}H) + D(Cl{-}Cl) - [D(C{-}Cl) + D(H{-}Cl)]$$
$$= (413\ \text{kJ} + 242\ \text{kJ}) - (328\ \text{kJ} + 431\ \text{kJ}) = -104\ \text{kJ}$$

Is this reaction exothermic or endothermic?

Calculate the enthalpy of reaction for:

$$CH_4(g) + Br_2(g) \rightarrow CH_3Br(g) + HBr(g)$$

Given:

TABLE 8.4 ■ Average Bond Enthalpies (kJ/mol)

Single Bonds

C—H	413	N—H	391	O—H	463	F—F	155
C—C	348	N—N	163	O—O	146		
C—N	293	N—O	201	O—F	190	Cl—F	253
C—O	358	N—F	272	O—Cl	203	Cl—Cl	242
C—F	485	N—Cl	200	O—I	234		
C—Cl	328	N—Br	243			Br—F	237
C—Br	276			S—H	339	Br—Cl	218
C—I	240	H—H	436	S—F	327	Br—Br	193
C—S	259	H—F	567	S—Cl	253		
		H—Cl	431	S—Br	218	I—Cl	208
Si—H	323	H—Br	366	S—S	266	I—Br	175
Si—Si	226	H—I	299			I—I	151
Si—C	301						
Si—O	368						
Si—Cl	464						

Covalent bond length

	C-C	C=C	C≡C
Bond length (Å)	1.54	1.34	1.2
Bond strength (kJ/mol)	348	614	839

As the number of bonds between two atoms increases, the bond grows shorter and stronger

Covalent bond length

TABLE 8.5 ■ Average Bond Lengths for Some Single, Double, and Triple Bonds

Bond	Bond Length (Å)	Bond	Bond Length (Å)
C—C	1.54	N—N	1.47
C=C	1.34	N=N	1.24
C≡C	1.20	N≡N	1.10
C—N	1.43	N—O	1.36
C=N	1.38	N=O	1.22
C≡N	1.16		
		O—O	1.48
C—O	1.43	O=O	1.21
C=O	1.23		
C≡O	1.13		

Key points

5. Drawing Lewis structure:
   - least electronegativity element in the middle
   - formal charge
6. Resonance structures: use when >2 Lewis structure can be written
7. Exceptions to the octet rule: odd number of e^-, more/less than octet
8. Bond strength and length:
   - stronger bond, shorter length,
   - use bond enthalpy to calculate reaction enthalpy

General Chemistry I

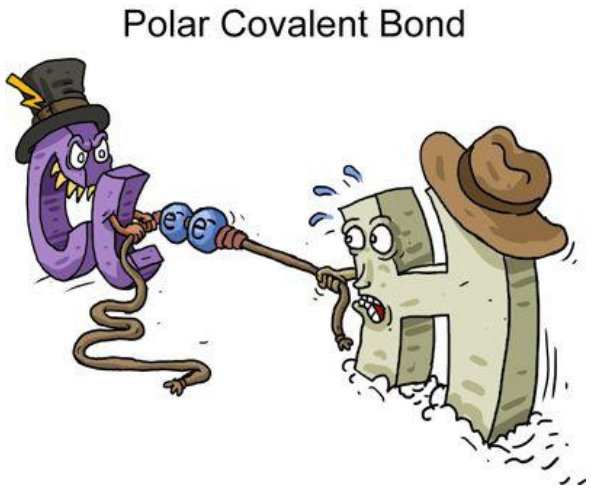
Chemistry: The central science

Phong D. Tran

University of Science and Technology of Hanoi

Email: tran-dinh.phong@usth.edu.vn

Lectures 9,10: Basic concept of chemical bonding



In which of following compounds the central atom(s) does (do) not obey the octet rule?

A. H_2S

B. H_2

C. NO

D. NaCl

Bonding within the NaCl crystal is:

A. Covalent

B. Ionic

C. Metallic

D. Coordination

Bonding within the diamond (an allotrope of carbon) is:

A. Covalent

B. Ionic

C. Metallic

D. Coordination

BF_3 is a:

A. Lewis acid

B. Lewis base

C. Polar compound

D. Non-polar compound

PF_3 is a:

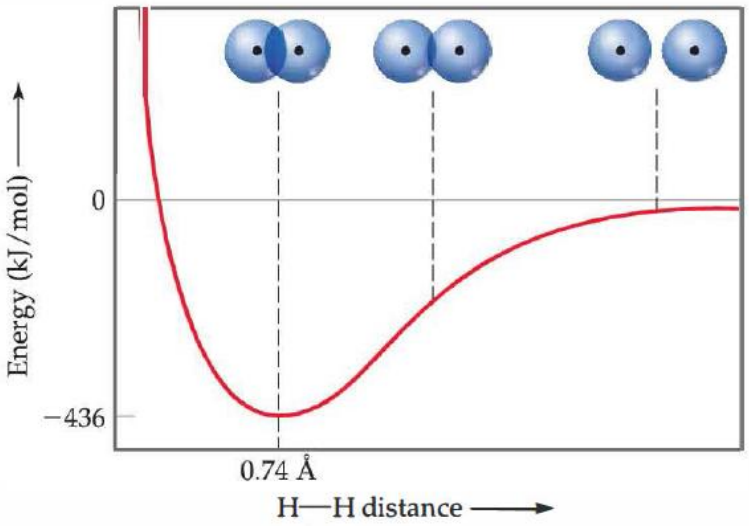
A. Lewis acid

B. Lewis base

C. Polar compound

D. Non-polar compound

From the following diagram, we can conclude:



A. H-H bond length is 0.74 Å

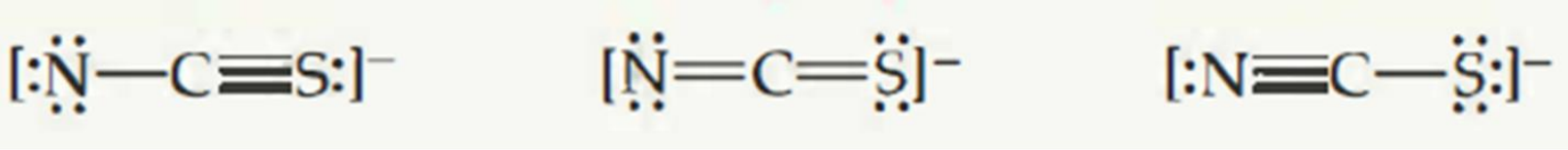
B. H_2 is in the most stable state (ground state) at H-H distance of 0.74 Å

C. H_2 is in the most active state at H-H distance of 0.74 Å

D. To destabilize and then eventually cleavage (cut) the H-H bond, we should elongate (extend) the H-H bond length.

Which of the following Lewis structures is the most likely one (*e.g.* based on an analysis of formal charge)?

Given electronegativity: C (2.5), N (3.0), and S (2.5).



(1) **(2)** **(3)**

A. (1)

B. (2)

C. (3)

Looking at the Lewis structure of SF_6 molecule, we can make following statement:

A. S (sulfur) atom obeys the octet rule

B. d-orbitals of S atoms should involve in the atomic orbital hybridization

C. S = F double bond is present within the SF_6 molecule

D. All 6 (six) S-F bonds are equal

Which of following molecules is (are) highly soluble in water at ambient conditions (room temperature and atmospheric pressure)?

A. O_2

B. CH_3CH_2OH

C. CH_3COOH

D. CS_2